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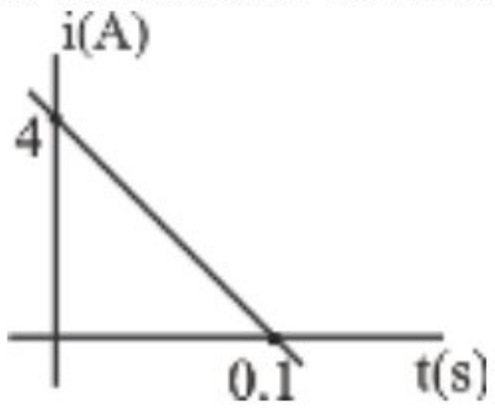
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PHYSICS

1) Some magnetic flux is changed from a coil of resistance 10Ω. As a result an induced current is developed in it which varies with time as shown in figure. Find the magnitude of change in flux through the coil in weber.



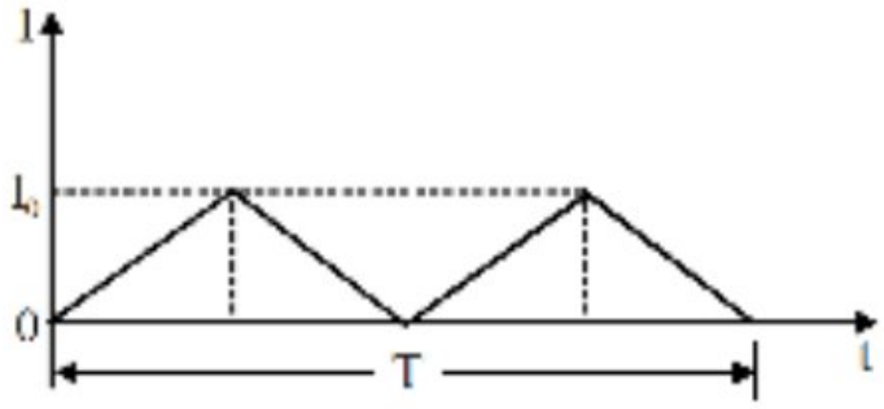
(1) 0.2 Wb

(2) 2 Wb

(3) 0.4 Wb

(4) 4 Wb

2) A current flowing through the inductor of inductance L varies as indicated in the Fig. The highest value of current is I_0. What will be the maximum voltage drop across the inductor :-



(1) zero

(2) $\frac{LI_0}{2T}$

(3) $\frac{LI_0}{\sqrt{2}T}$

(4) $4L\frac{I_0}{T}$

3) Magnetic flux ϕ (in weber) linked with a closed circuit of resistance 10 ohm varies with time t (in seconds) as $\phi = 4t^2 - 4t + 2$. The induced emf (in volt) in the circuit at t = 1. sec is :

(1) 1

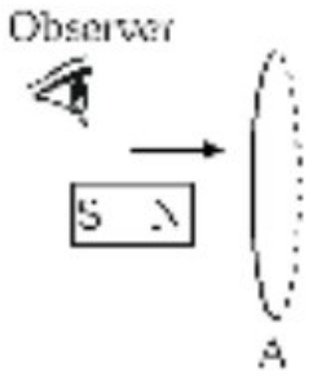
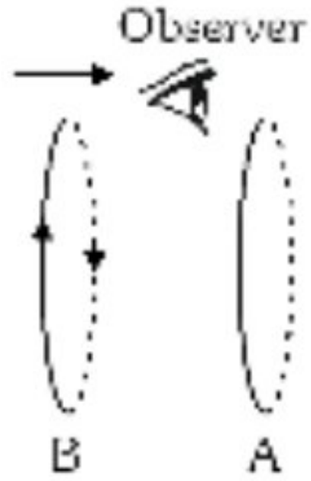
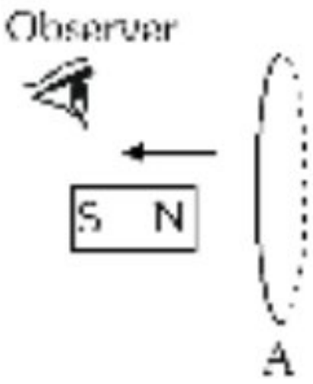
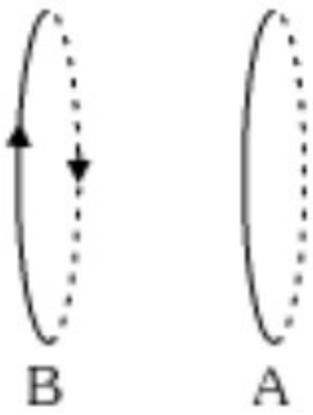
(2) 2

(3) 3

(4) 4

4) Match the following lists.

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	List-I		List-II (Direction of induce current in loop A)
(P)		(1)	Clockwise
(Q)		(2)	Anticlockwise
(R)		(3)	Zero
(S)		(4)	Cannot determine

(1) P-1, Q-2, R-3, S-4

(2) P-2, Q-1, R-1, S-3

(3) P-3, Q-4, R-1, S-4

(4) P-4, Q-3, R-1, S-2

5) **Statement-1 :** When a magnet is made to fall freely through a closed coil, its acceleration is always less than acceleration due to gravity.

and

Statement-2 : Current induced in the coil opposes the motion of the magnet, as per Lenz's law.

(1) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.

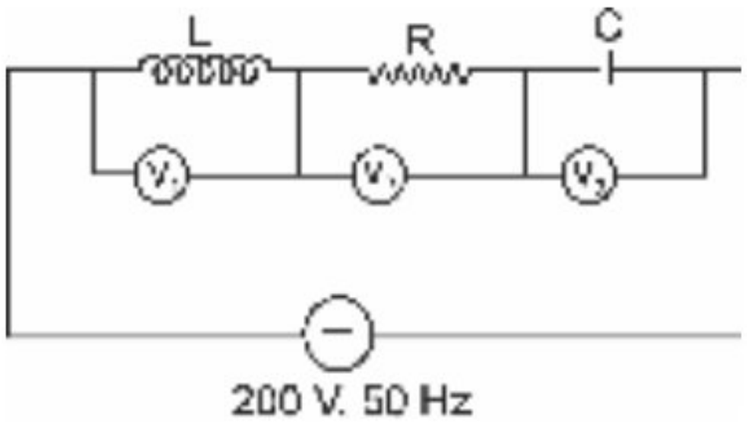
(2) Statement-1 is true, Statement-2 is true, Statement-2 is NOT a correct explanation for Statement-1.

(3) Statement-1 is True, Statement-2 is False.

(4) Statement-1 is False, Statement-2 is True.

6) If the readings of v_1 and v_3 are 100 volt each then reading of v_2 is :

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(1) Zero

(2) 100 volt

(3) 200 volt

(4) cannot be determined by given information.

7) A 10 metre wire kept in east-west falling with velocity 5 m/sec perpendicular to the field 0.3 × 10^{-4} Wb/m^2. The induced e.m.f. across the terminal will be :

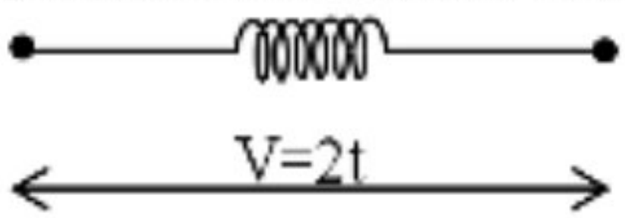
(1) 0.15 V

(2) 1.5 mV

(3) 1.5 V

(4) 15.0 V

8) A variable voltage V = 2t is applied across an inductor of inductance L = 2 Henry as shown in figure. Choose incorrect option.



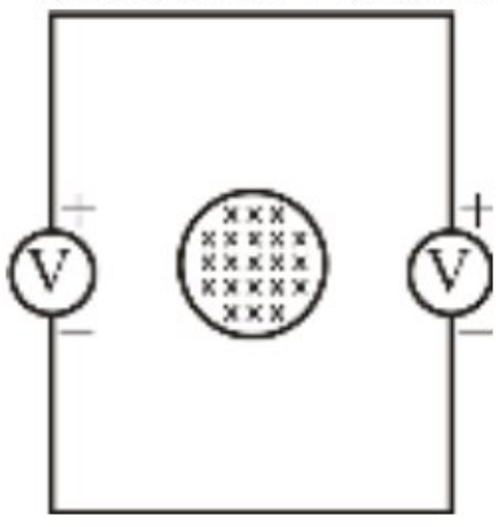
(1) current versus time graph is a parabola

(2) energy stored in magnetic field at t = 2s is 4 joule

(3) potential energy at time t = 1 s in magnetic field is increasing at a rate of 1 J/s

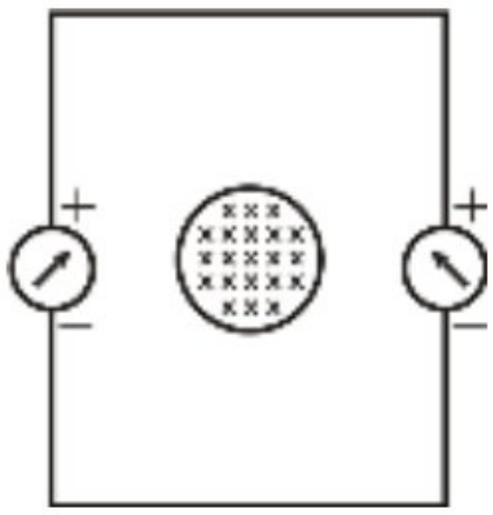
(4) energy stored in magnetic field is zero all time

9) Two similar voltmeters with large resistance are connected in a loop as shown. Inside the loop, there is a cylindrical region through which magnetic flux is distributed uniformly. The magnetic field increases uniformly with time. The voltmeters are bidirectional i.e. the needle deflects to right if current in it flows from + terminal to – terminal and needle deflects to left if current in it flows from – terminal to + terminal. Which of the following figures shows the readings of voltmeters correctly?

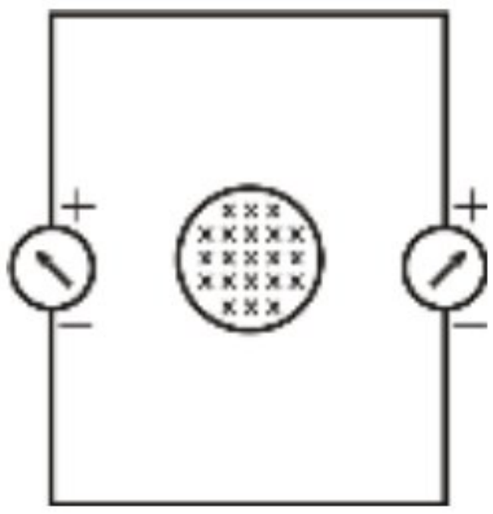


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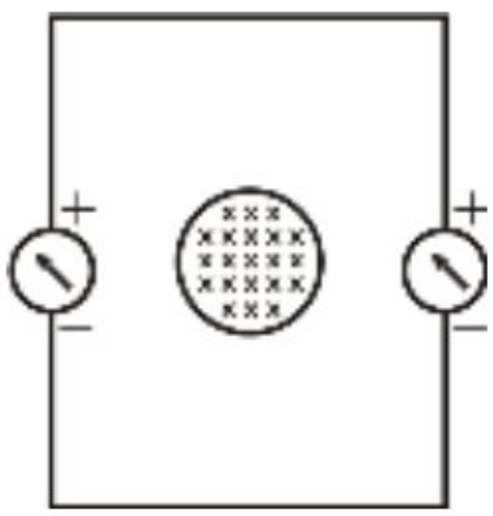
(1)



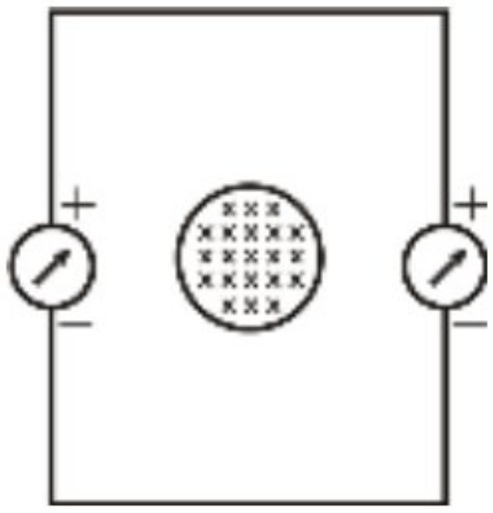
(2)



(3)



(4)



10) Primary side of a transformer is connected to 230 V, 50 Hz supply. Turns ratio of primary to secondary winding is 10 : 1. Load resistance connected to secondary side is 46 Ω. The power consumed in it is :

(1) 12.5 W

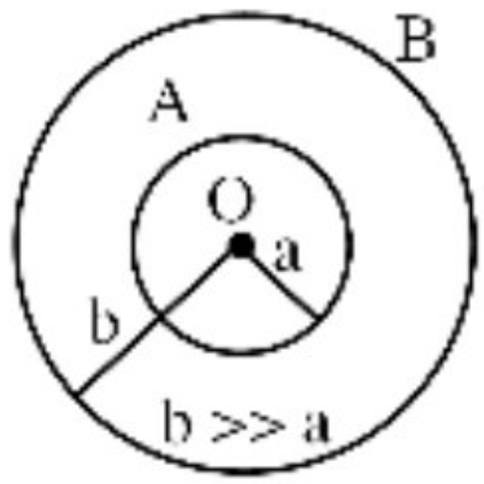
(2) 10.0 W

(3) 11.5 W

(4) 12.0 W

11) Two conducting circular loops A and B are placed in the same plane with their centres

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coinciding as shown in figure. The mutual inductance between them is:

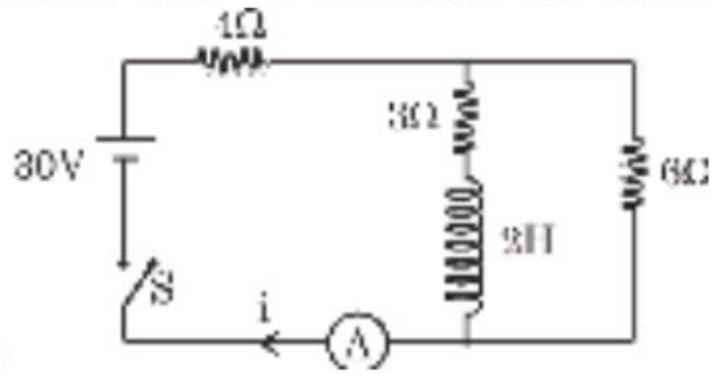
(1) $\frac{\mu_0 \pi a^2}{2b}$

(2) $\frac{\mu_0}{2\pi} \cdot \frac{b^2}{a}$

(3) $\frac{\mu_0 \pi b^2}{2a}$

(4) $\frac{\mu_0}{2\pi} \cdot \frac{a^2}{b}$

12) Switch is closed at t = 0 then find the value of current i as shown in diagram at t = 0 and t = ∞



respectively :-

(1) 5A, 3A

(2) 3A, 5A

(3) 3A, 4A

(4) 5A, 4A

13) A square wave current switching rapidly between + 0.5 ampere and –0.5 ampere is passed through an A.C ammeter. Then the reading shown by it, is

(1) 0.25 ampere

(2) 0.5 ampere

(3) $\frac{0.5}{\sqrt{2}}$ **ampere**

(4) $0.5 \times \sqrt{2}$ **ampere**

14) In an a. c. circuit, the instantaneous e.m.f. and current are given by

e = 100 sin 30 t

i = 20 sin $\left(30t - \frac{\pi}{4}\right)$

In one cycle of a.c., the average power consumed by the circuit and the wattless current are, respectively.

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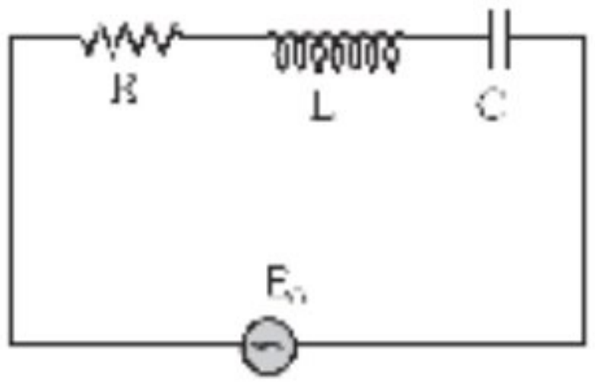
(1) $\frac{1000}{\sqrt{2}}$, 10

(2) $\frac{50}{\sqrt{2}}$, 0

(3) 50, 0

(4) 50, 10

15) If E_0 = 200 volt, R = 25 ohm, L = 0.1 H and C = 10^{-5} F and the frequency is variable, then the current at f = 0 and f = ∞ will be respectively -



(1) 0 A, 8 A

(2) 8 A, 0 A

(3) 8A, 8A

(4) 0A, 0A

16) The r.m.s. value in time T of potential difference V shown in the figure is :-



(1) $\frac{V_0}{\sqrt{3}}$

(2) V_0

(3) $\frac{V_0}{\sqrt{2}}$

(4) $\frac{V_0}{2}$

17) An R-L circuit having 100 Ω resistor and 1 Henry inductor are connected in series combination with an A.C. source $V = 10\sin(100t)$ then calculate power factor of the circuit.

(1) $\frac{1}{2}$

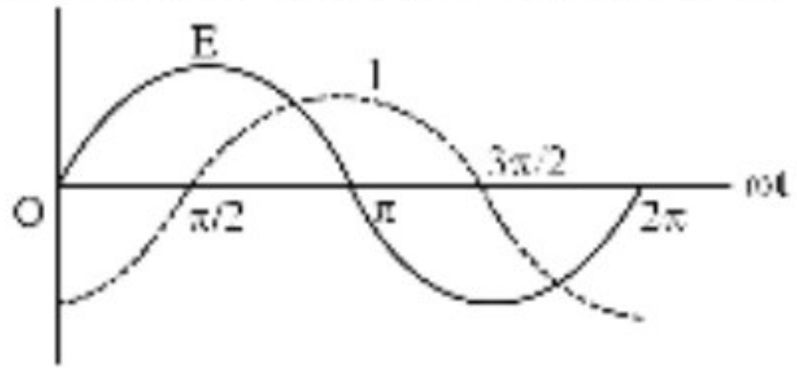
(2) $\frac{1}{\sqrt{2}}$

(3) $\frac{\sqrt{3}}{2}$

(4) 1

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18) The variation of the instantaneous current (I) and the instantaneous emf (E) in a circuit is as shown in fig. Which of the following statements is correct –



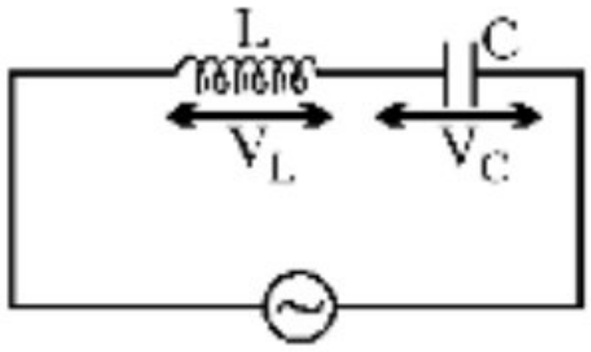
(1) The voltage lags behind the current by π/2

(2) The voltage leads the current by π/2

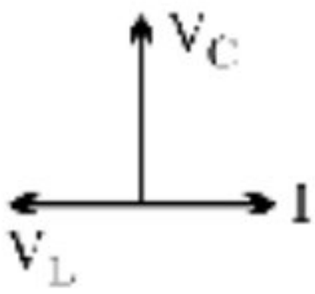
(3) The voltage and the current are in phase

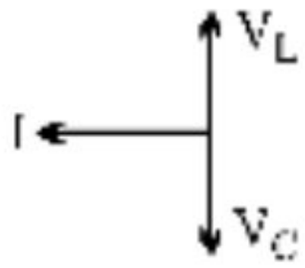
(4) The voltage leads the current by π

19) The current I, potential difference V_L across the inductor and potential difference V_C across the capacitor in circuit as shown in the figure are best represented vectorially as

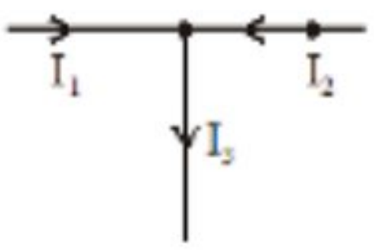


(1)

(2) 

(3) 

(4)



20) If current $I_1 = 3A \sin \omega t$ and $I_2 = 4A \cos \omega t$, then I_3 is :-

(1) $5A \sin (\omega t + 53°)$

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(2) 5A sin (ωt + 37°)

(3) 5A sin (ωt + 45°)

(4) 5A sin (ωt + 30°)

21) In an electromagnetic wave, at an instant and at a particular position, the electric field is along the negative z-axis and magnetic field is along the positive x-axis. Then the direction of propagation of electromagnetic wave is :

(1) at 45º angle from positive y-axis

(2) negative y-axis

(3) positive z-axis

(4) positive y-axis

22) The electric field associated with an electromagnetic wave in vacuum is given by $\vec{E} = 40 \cos\left(kz - 6 \times 10^8 t\right) \hat{i}$, where E, z & t are in volt/m, meter & second respectively. The value of propagation constant k is :-

(1) 6 m^{-1}

(2) 3 m^{-1}

(3) 2 m^{-1}

(4) 0.5 m^{-1}

23) The magnetic field in the plane electromagnetic wave is given by $B_z = 2 \times 10^{-7} \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t)$ tesla. The expression for electric field will be:

(1) $E_z = 30\sqrt{2} \sin\left(0.5 \times 10^3 x + 1.5 \times 10^{11} t\right)$ V/m

(2) $E_z = 60 \sin\left(0.5 \times 10^3 x + 1.5 \times 10^{11} t\right)$ V/m

(3) $E_y = -30\sqrt{2} \sin\left(0.5 \times 10^{11} x + 1.5 \times 10^3 t\right)$ V/m

(4) $E_y = -60 \sin\left(0.5 \times 10^3 x + 1.5 \times 10^{11} t\right)$ V/m

24) In the given electromagnetic wave $E_y = 600 \sin(\omega t - kx)$ Vm^{-1}, intensity of the associated light beam is (in W/m^2); (Given $\epsilon_0 = 9 \times 10^{-12} C^2 N^{-1} m^{-2}$)

(1) 486

(2) 243

(3) 729

(4) 972

25) Match **List-I** with **List-II** :-

	List-I EM-Wave		**List-II Wavelength Range**
(A)	Infra-red	(I)	< 10^{-3} nm

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(B)	Ultraviolet	(II)	400 nm to 1 nm
(C)	X-rays	(III)	1 mm to 700 nm
(D)	Gamma rays	(IV)	1 nm to 10^{-3} nm

Choose the correct answer from the options given below :

(1) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

(2) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

(3) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(4) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)

26) Electromagnetic waves travel in a medium with speed of 1.5×10^8 ms^{-1}. The relative permeability of the medium is 2.0. The relative permittivity will be :

(1) 5

(2) 1

(3) 4

(4) 2

27) In an EM wave propagating along X-direction magnetic field oscillates at a frequency of 3×10^{10} Hz. The magnetic field oscillates along Y-direction and has an amplitude of 10^{-7} T. The expression for electric field will be :-

(1) $E_z = 30 \sin 2\pi(100x - 3\times10^{10} t)$ V/m

(2) $E_z = 300 \sin 2\pi(100x - 3\times10^{10} t)$ V/m

(3) $E_y = 30 \sin 2\pi(100x - 3\times10^{10} t)$ V/m

(4) $E_y = 300 \sin 2\pi(100x - 3\times10^{10} t)$ V/m

28) Cu and Ge are cooled to 70 K then :–

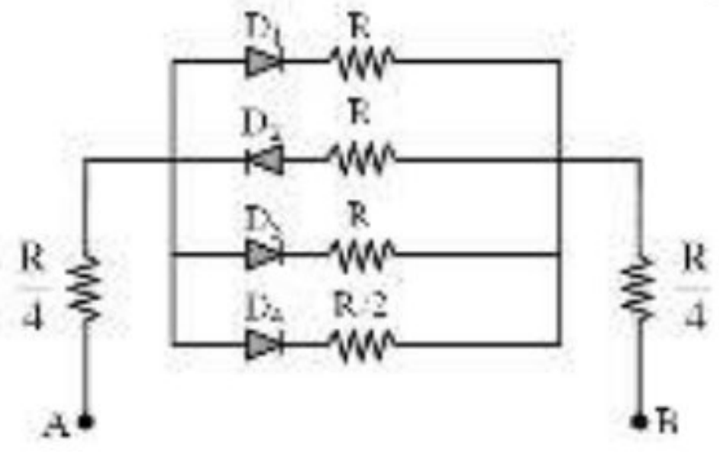
(1) the resistance of Cu will increase and that of Ge will decrease

(2) the resistance of Cu will decrease and that of Ge will increase

(3) the resistance of both Cu and Ge decrease

(4) the resistance of both Cu and Ge increase

29) In the following circuits PN - junction diodes D_1, D_2 and D_3 are ideal for the following potentials of A and B. The correct increasing order of resistance between A and B will be



(i) – 10V, – 5V (ii) – 5V, – 10V (iii) – 4V, – 12V

(1) (i) < (ii) < (iii)

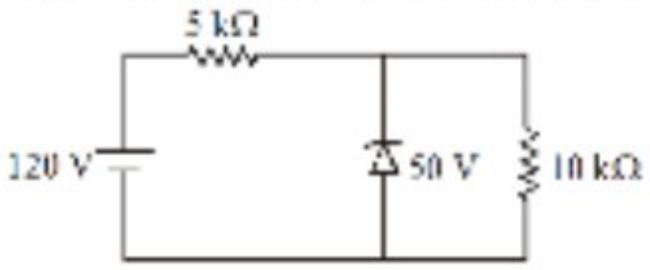
(2) (iii) < (ii) < (i)

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(3) (ii) = (iii) < (i)

(4) (i) = (iii) < (ii)

30) For the circuit shown below, the current through the Zener diode is :



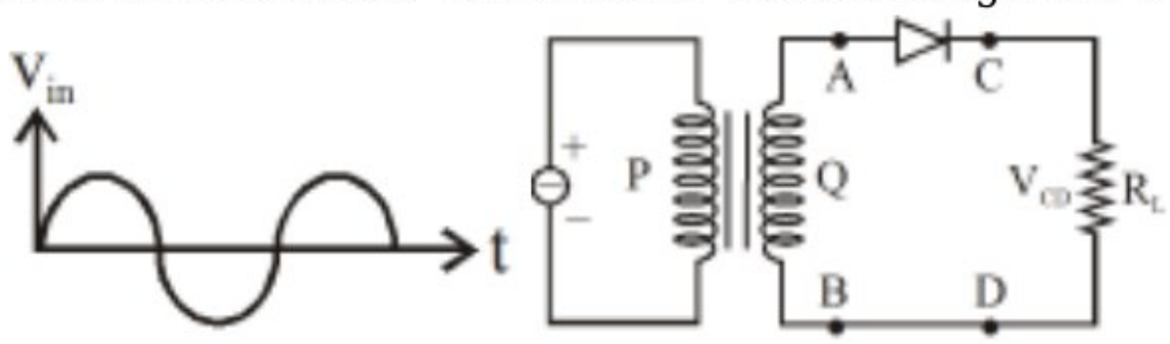
(1) 5 mA

(2) Zero

(3) 14 mA

(4) 9 mA

31) In the half-wave rectifier circuit shown. Which one one of the following wave forms is true for V_{CD}, if the input is as shown ?



(1)



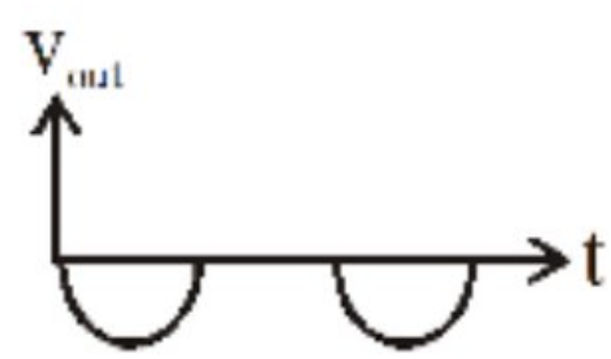
(2)



(3)

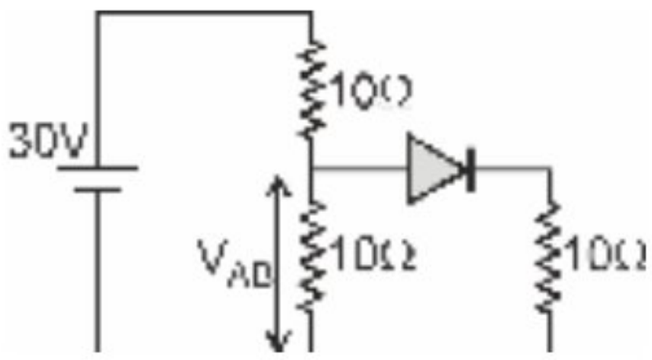


(4)



32) An ideal cell of emf 30 V is connected across three resistance of 10 Ω each and an ideal diode as shown. Then the magnitude of potential difference V_{AB} across at 10 Ω resistor as shown is :

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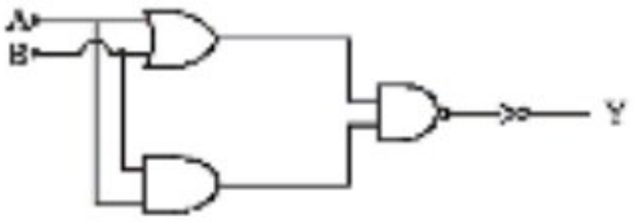
(1) 10 V

(2) 20 V

(3) 30 V

(4) None of these

33) The logic operations performed by the given digital circuit is equivalent to:



(1) AND

(2) NOR

(3) OR

(4) NAND

34) Pure Si at 300 K have equal electron (n_e) and hole (n_h) concentrations of 1.5×10^{16} m^{-3}. Doping by indium increases n_h to 4.5×10^{22} m^{-3}. Calculate n_e in the doped Si.

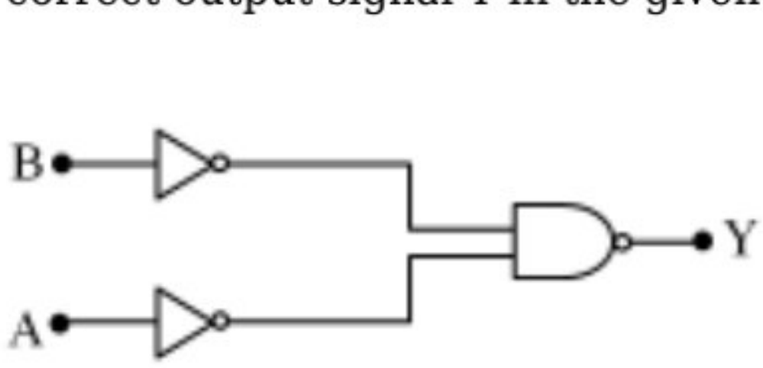
(1) (5×10^{9} m^{-3})

(2) (5×10^{8} m^{-3})

(3) (2×10^{9} m^{-3})

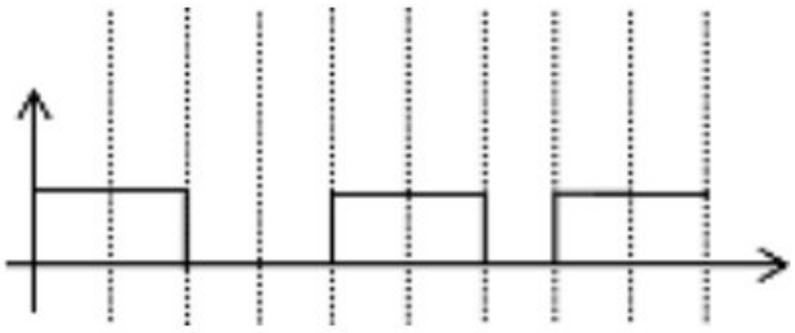
(4) (8×10^{9} m^{-3})

35) Identify the correct output signal Y in the given combination of gates (as shown) for the given inputs A and B.



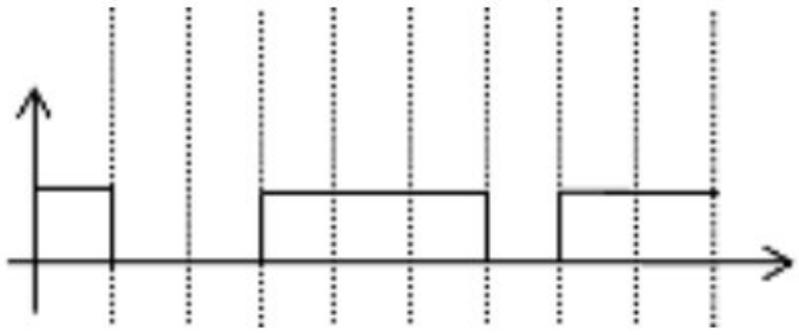


(1)



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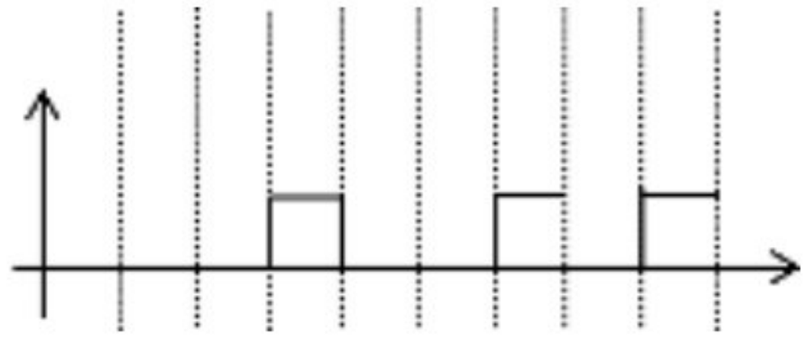
(2)



(3)



(4)



36) In the photoelectric experiment, fastest electron has speed v if wavelength used is λ. If wavelength used is $\frac{3\lambda}{4}$, the speed of fastest electron will be :-

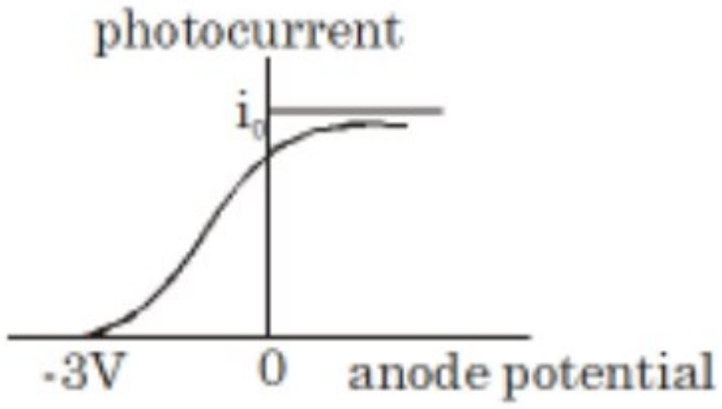
(1) $\left(\frac{3}{4}\right)^{\frac{1}{2}} v$

(2) $\left(\frac{4}{3}\right)^{\frac{1}{2}} v$

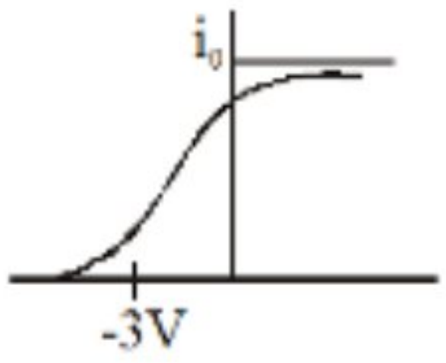
(3) Less than $\left(\frac{4}{3}\right)^{\frac{1}{2}} v$

(4) Greater than $\left(\frac{4}{3}\right)^{\frac{1}{2}} v$

37) Figure shows the anode potential Vs photo-current graph when photons of 6 eV are incident on cathode in a photoelectric experiment set-up. In the same experimental set-up, if photons of 8 eV and same intensity are incident on cathode, then anode-potential vs photo-current graph will be (assume 100% efficiency of photons to eject photo-electrons in both cases)

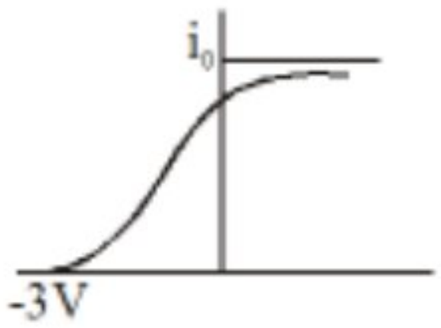


(1)

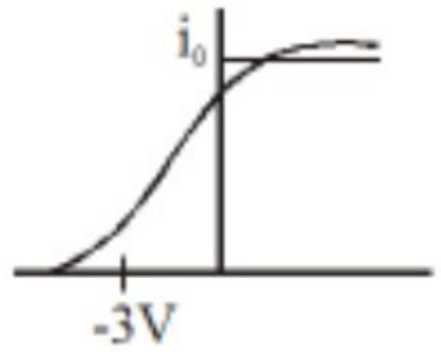


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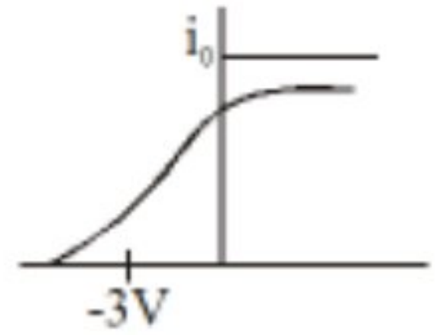
(2)



(3)

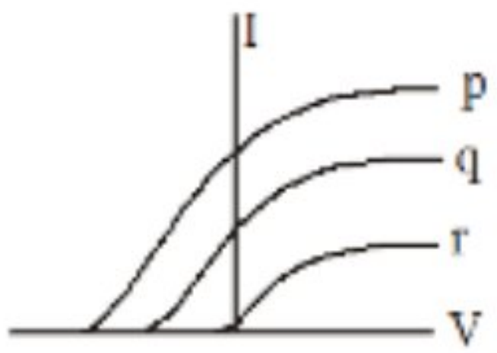


(4)

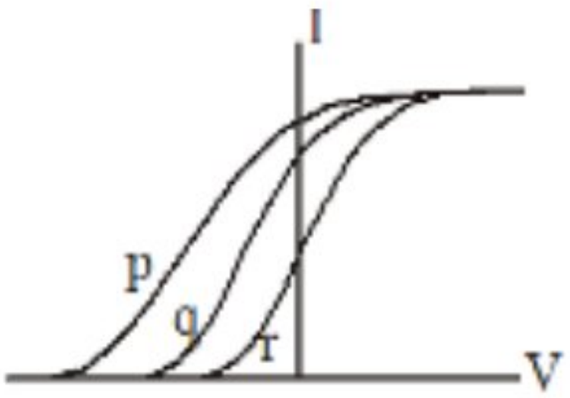


38) Photoelectric effect experiments are performed using three different metal plates p, q and r having work functions ϕ_p = 2.0 eV, ϕ_q = 2.5 eV and ϕ_r = 3.0 eV, respectively. A light beam containing wavelengths of 550 nm, 450 nm and 350 nm with equal intensities illuminates each of the plates. The correct I-V graph for the experiment is :

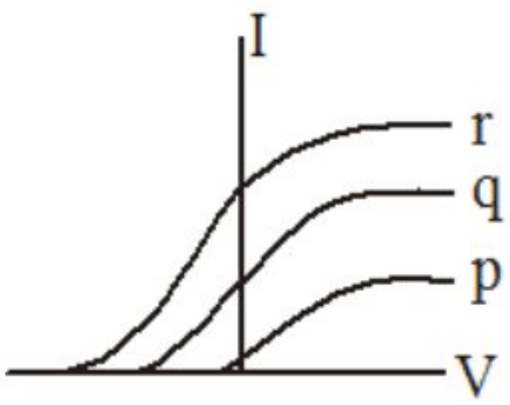
(1)



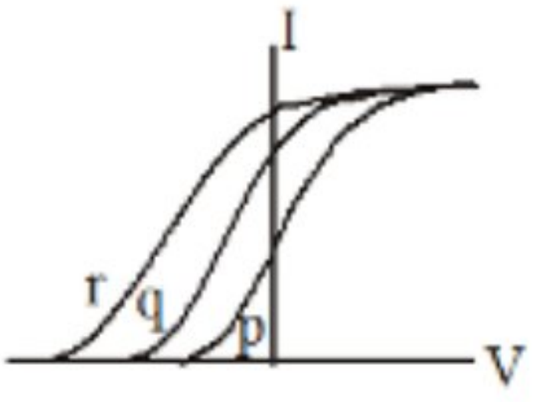
(2)



(3)



(4)



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39) In Bohr's atomic model for hydrogen like atoms match the following column :

Column I		Column II	
(A)	If electron jumps from n = 2 to n = 1	(p)	Speed of electron will become 2 times
(B)	If electron jumps from n = 1 to n = 4	(q)	Kinetic energy of electron will become 4 times
(C)	If electron jumps from n = 4 to n = 1	(r)	Angular momentum of electron will become 2 times
		(s)	Angular velocity of electron will become 4 times
		(t)	None

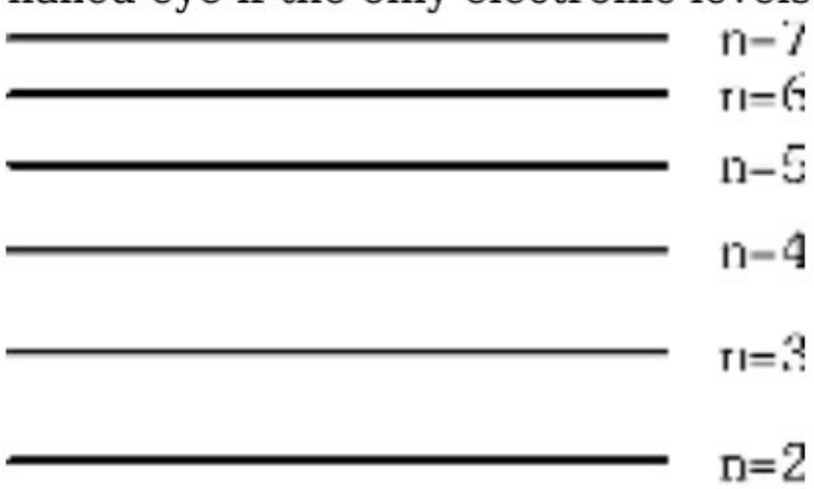
(1) A-p; B-r; C-t

(2) A-p,q; B-t, C-t

(3) A-r,p; B-s,t; C-p

(4) A-p,q; B-r, C-t

40) The maximum number of emission lines for atomic hydrogen that you would expect to see with naked eye if the only electronic levels involved are those shown in the figure, is



(1) 6

(2) 5

(3) 21

(4) ∞

41) In a radioactive decay, neither the atomic number nor the mass number changes. Which of the following particles is emitted in the decay?

(1) proton

(2) neutron

(3) electron

(4) photon.

42) The binding energy per nucleon for C^{12} is 7.68 MeV and that for C^{13} is 7.5 MeV. The energy required to remove a neutron from C^{13} is

(1) 5.34 MeV

(2) 5.5 MeV

(3) 9.5 MeV

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(4) 9.34 MeV

43) Consider the following nuclear reactions and select the incorrect statement from the options that follow.

Reaction I : $n \rightarrow p + e^- + \bar{\nu}$

Reaction II : $p \rightarrow n + e^+ + \nu$

(1) Free neutron is unstable, therefore reaction I is possible

(2) Free proton is stable, therefore reaction II is not possible

(3) Inside a nucleus, both decays (reaction I and II) are possible

(4) Inside a nucleus, reaction I is not possible but reaction II is possible.

44) The following masses are known

$^{1}_{0}n$ 1.008665u

$^{1}_{1}H$ 1.007825u

$^{62}_{28}Ni$ 61.928349u

The binding energy of $^{62}_{28}Ni$, in MeV, is closest to :

(1) 545

(2) 600

(3) 650

(4) 710

45) Two particles move at right angle to each other. Their de-Broglie wavelengths are λ_1 and λ_2 respectively. The particles suffer perfectly inelastic collision. The de-Broglie wavelength λ, of the final particle, is given by :

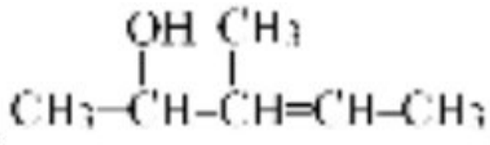
(1) $\lambda = \frac{\lambda_1 + \lambda_2}{2}$

(2) $\frac{2}{\lambda} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$

(3) $\lambda = \sqrt{\lambda_1 \lambda_2}$

(4) $\frac{1}{\lambda^2} = \frac{1}{\lambda_1^2} + \frac{1}{\lambda_2^2}$

CHEMISTRY

1) The IUPAC name of the compound  is :-

(1) 3, 4-Dimethylbut-2-en-4-ol

(2) 3-Methylpent-2-en-4-ol

(3) 3-Methylpent-3-en-2-ol

(4) 1, 2-Dimethylbut-2-en-1-ol

2)

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Which of the following will give two isomeric monochloro derivatives (structural only) :-

(1) 2,3-Dimethyl butane

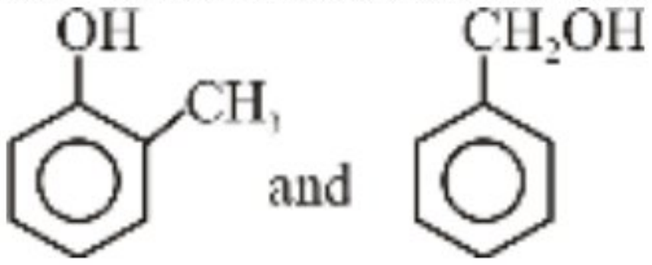
(2) n-Hexane

(3) Isopentane

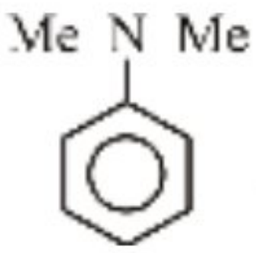
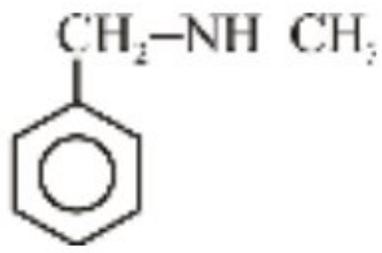
(4) Ethane

3)

Number of correct statement in following :

(a)  and

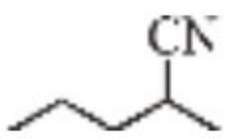
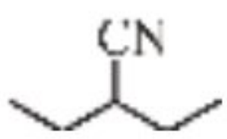
Functional group isomers

(b)  and 

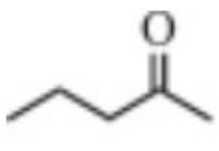
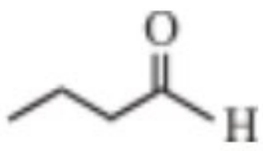
Metamers

(c) CH_3-C≡C-CH_3 and 

Ring chain isomer

(d)  and 

Chain isomers

(e)  and 

Functional group isomers

(1) 2

(2) 5

(3) 4

(4) 3

4) How many structural isomeric primary alcohols are possible for $C_5H_{11}OH$:-

(1) 8

(2) 5

(3) 2

(4) 4

5)

Identify correct statement :

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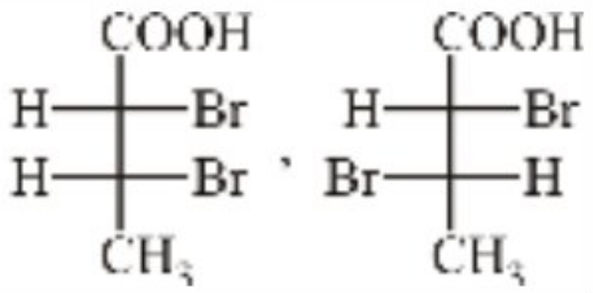
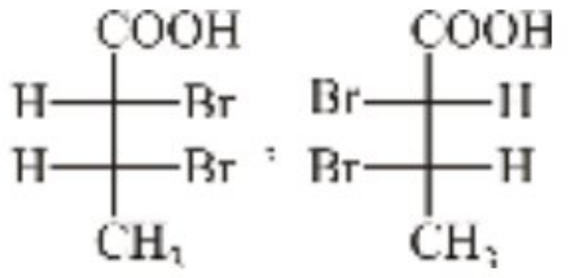
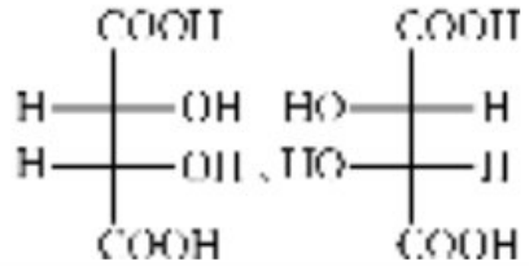
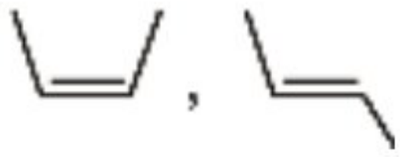
(1) Compound having D-configuration is always dextrorotatory in nature.

(2) isobutane does not have any chiral centre

(3) Geometrical isomers have same boiling point

(4) Enantiomers have different melting point

6) Which pair (List-I) having incorrect relationship (List-II).

	List-I	List-II
(1)		Diastereomer
(2)		Enantiomer
(3)		Enantiomer
(4)		Diastereomer

(1) 1

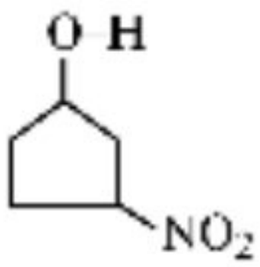
(2) 2

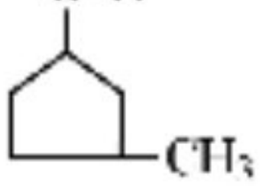
(3) 3

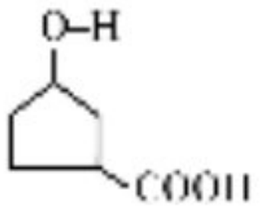
(4) 4

7)

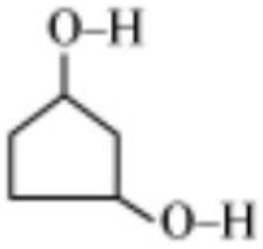
Which is most acidic compound ?

(1) 

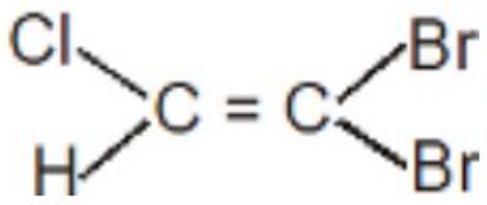
(2) 

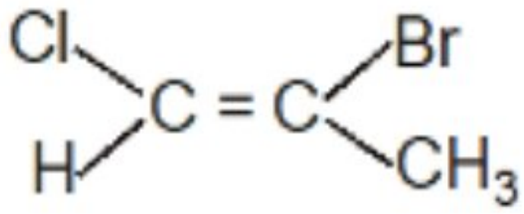
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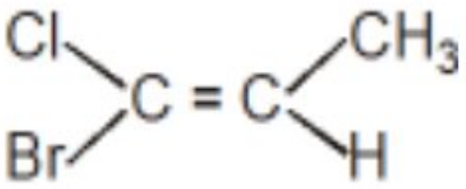
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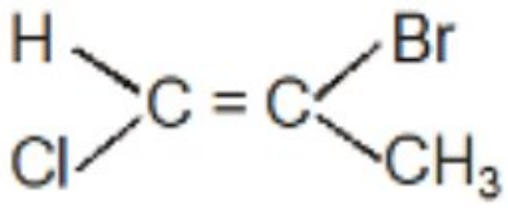
(4) 

8) Which is a pair of geometrical isomers ?

I. 

II. 

III. 

IV. 

(1) I and II

(2) I and III

(3) II and IV

(4) III and IV

9) Which is incorrect statement :

(1) Cellulose is linear polymer of α-D-glucose

(2) Sucrose is composed by α-D-glucose and β-D-fructose

(3) α-D-glucose and β-D-glucose are anomers

(4) Glucose and mannose are epimer

10) Maltose is formed by :-

(1) α-D-glucose and α-D-glucose

(2) α-D-glucose and β-D-glucose

(3) α-D-glucose and β-D-fructose

(4) α-D-glucose and β-D-galactose

11) Which of the following is non-reducing sugar :-

(1) Glucose

(2) Fructose

(3) Sucrose

(4) Lactose

12) Which of the following statements is false ?

(1) All monosaccharides whether aldose or ketose are reducing sugars.

(2) Both starch and cellulose are polymer of glucose

(3) During denaturation 1° and 2° protein structure get destroyed but 3º structure remain the same.

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(4) In fibrous protein chain are held together by hydrogen bond and disulphide bond.

13)

One letter code for Glycine & lysine are respectively ?

(1) G, Y
(2) G, L
(3) G, K
(4) C, S

14) What is compound Z?

$$CH_3\text{-}CH_2\text{-}CH_2\text{-}Br \xrightarrow{NaCN} X \xrightarrow[\Delta]{H_3O^{\oplus}} Y \xrightarrow[H^{\oplus}]{CH_3\text{-}CH_2\text{-}OH} Z$$

(1) $CH_3\text{-}CH\text{=}CHC(=O)OH$

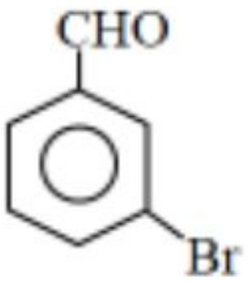
(2) $CH_3\text{-}CH_2\text{-}CH_2CH\text{=}NOCH_2CH_3$

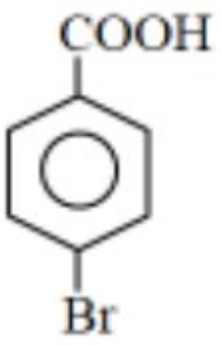
(3) $CH_3\text{-}CH_2\text{-}CH_2CH(OCH_2CH_3)_2$

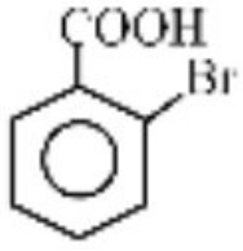
(4) $CH_3\text{-}CH_2\text{-}CH_2\text{-}C(=O)\text{-}OCH_2CH_3$

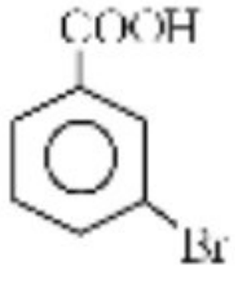
15) $C_6H_6 \xrightarrow[AlCl_3]{Cl_2} A \xrightarrow[\text{Dry ether}]{Mg} B \xrightarrow[2.\ H_3O^+]{1.\ CO_2} C \xrightarrow[FeBr_3]{Br_2} D$,

Last product D will be :

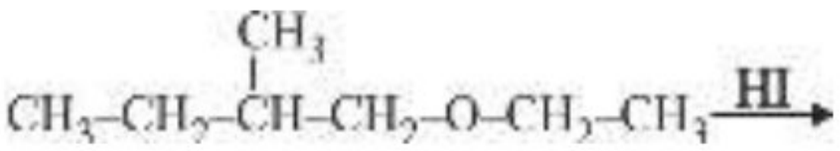
(1) 

(2) 

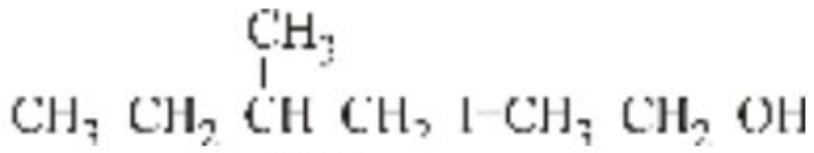
(3) 

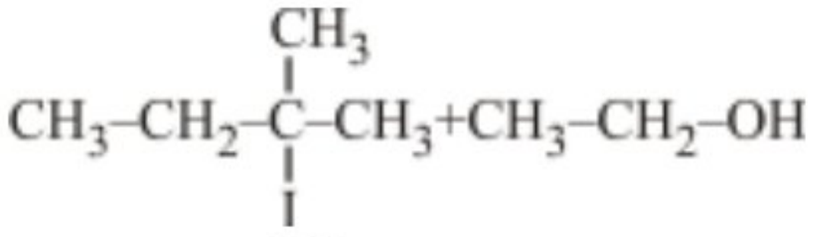
(4) 

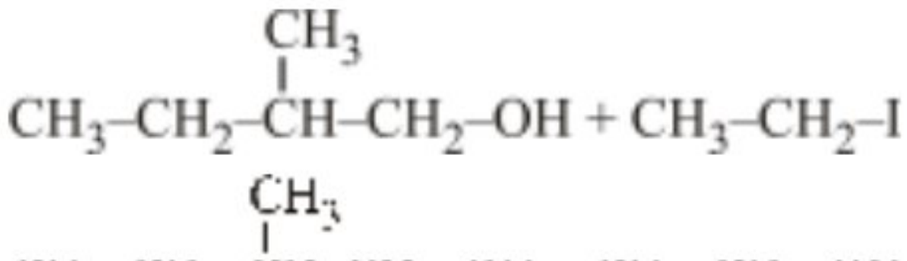
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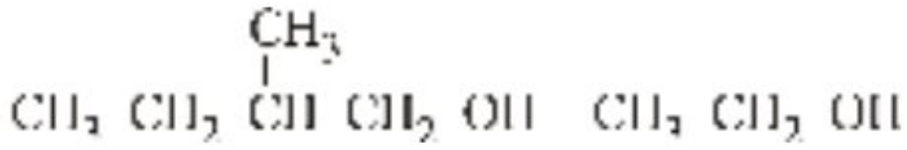
16) 

Product are :-

(1) 

(2) 

(3) 

(4) 

17) Incorrect statement about Aspirine.

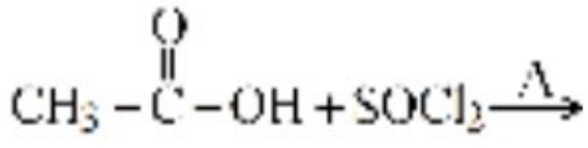
(1) Contains phenolic group

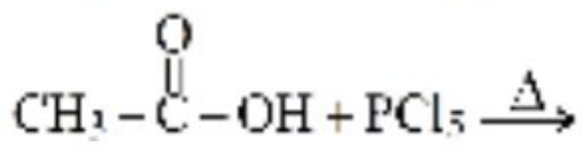
(2) Contain ester group

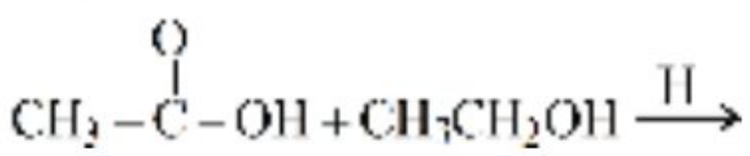
(3) Contains carboxylic group

(4) It is used as antipyretic and analgesic

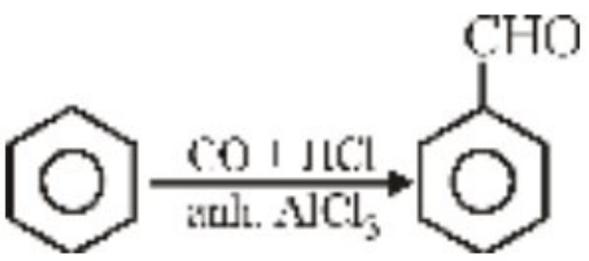
18) Which of the following reaction does not involve cleavage of oxygen carbon bond of carboxylic acid :-

(1) 

(2) 

(3) 

(4) 

19) 

The reaction is:-

(1) Gattermann-Koch reaction

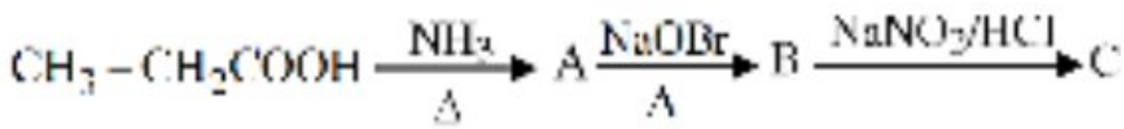
(2) Gattermann aldehyde synthesis

(3) Gattermann reaction

(4) Etard reaction

20) Give the name of A, B and C in the following reaction

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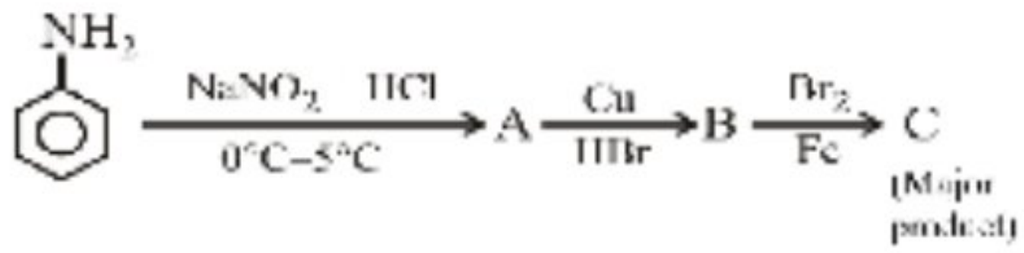
	A	B	C
(1)	Acetamide	Methyl Amine	Ethanol
(2)	Methyl amine	Acetamide	Ethanol
(3)	Propanamide	Ethylamine	Ethyl alcohol
(4)	Acetamide	Methyl amine	Diazonium chloride

(1) 1

(2) 2

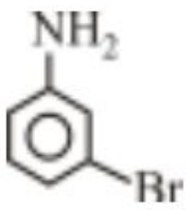
(3) 3

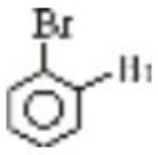
(4) 4



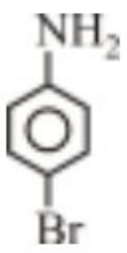
21)

Identify end product C in above sequence.

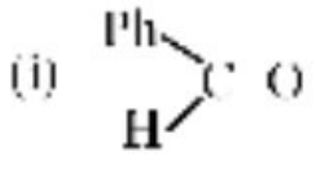
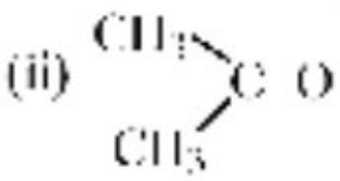
(1) 

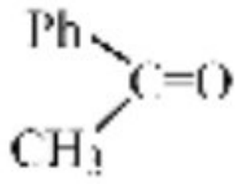
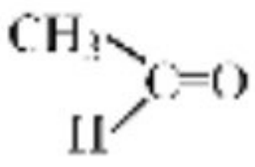
(2) 

(3) 

(4) 

22) Give decreasing order of reactivity for nucleophilic addition reaction :-

(i)  (ii) 

(iii)  (iv) 

(1) (i) > (ii) > (iii) > (iv)

(2) (iv) > (i) > (ii) > (iii)

(3) (iv) > (i) > (iii) > (ii)

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(4) (ii) > (i) > (iv) > (iii)

23) The reaction

$$C_6H_5CHO + CH_3CHO \xrightarrow[\Delta]{NaOH} C_6H_5CH=CHCHO$$

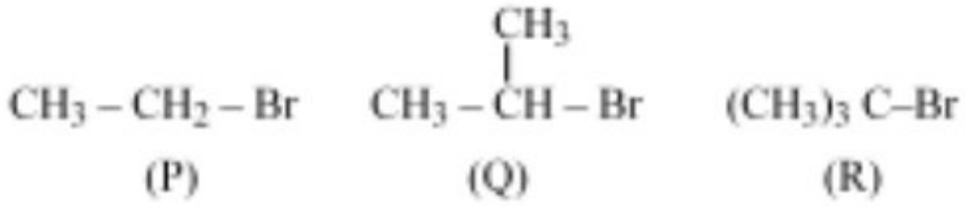
is known as :-

(1) Cross cannizzaro reaction

(2) Cross Aldol Condensation

(3) HVZ reaction

(4) Cannizzaro reaction

24) 

(P) (Q) (R)

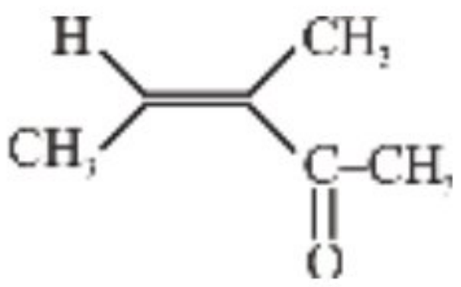
Correct reactivity order for williamson ether synthesis ?

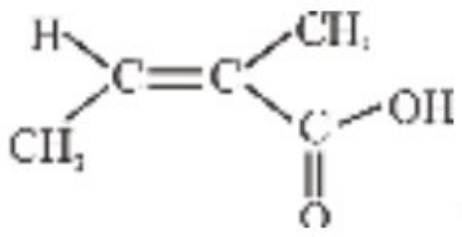
(1) R > Q > P

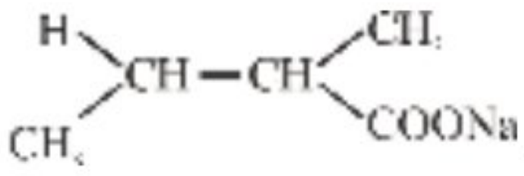
(2) P > Q > R

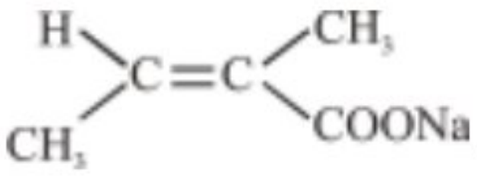
(3) Q > P > R

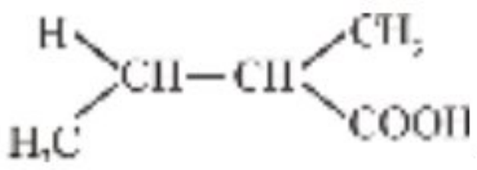
(4) P > R > Q

25) Identify the product :-  + NaOCl →

(1)  + NaCl

(2)  + $CHCl_3$

(3)  + $CHCl_3$

(4)  + NaCl

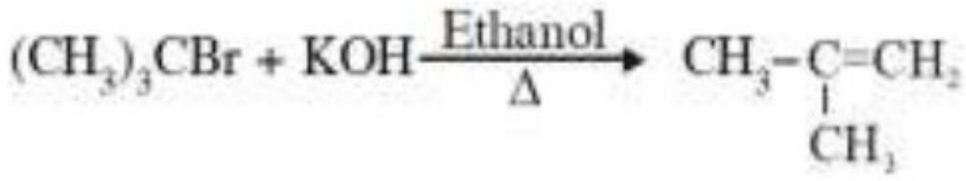
26) Which is not correct regarding major product?

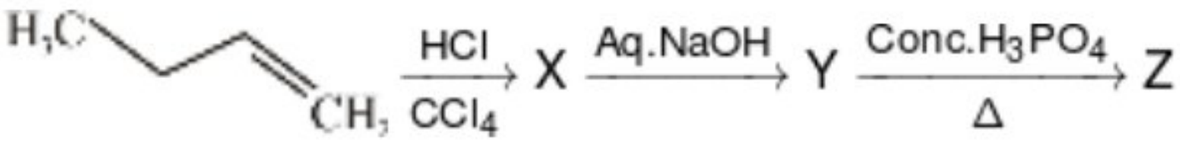
(1) $CH_3\text{–}CH_2\text{–}CH_2\text{–}Cl \xrightarrow[\text{Acetone}]{NaI} CH_3\text{–}CH_2\text{–}CH_2\text{–}I$

(2) $CH_3\text{–}CH{=}CH_2 \xrightarrow[\text{Peroxide}]{HBr} CH_3\text{–}CH_2\text{–}CH_2\text{–}Br$

(3) $Ph\text{–}ONa + C_2H_5Cl \rightarrow Ph\text{–}OH + CH_2{=}CH_2$

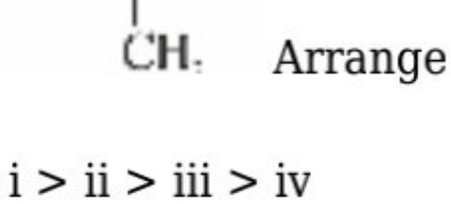
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(4) 

27)  Major product is :

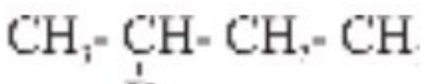
(1) $CH_3–CH_2–CH=CH_2$

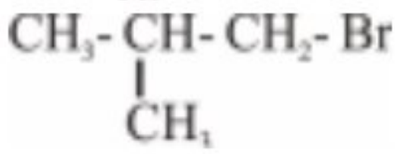
(2) $CH_3–CH=CH–CH_3$

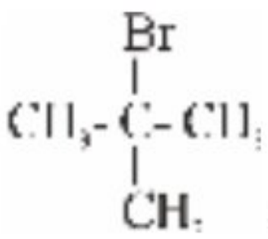
(3) 

(4) $CH_2=CH–CH=CH_2$

28) (i) $CH_3–CH_2–CH_2–CH_2–Br$

(ii) 

(iii) 

(iv)  Arrange the above compounds according to reactivity towards S_N1 reaction

(1) i > ii > iii > iv

(2) iv > iii > ii > i

(3) iv > ii > i > iii

(4) iv > ii > iii > i

29) Reaction of phenol with carbon dioxide in presence of sodium Hydroxide followed by acidification introduce which functional group ?

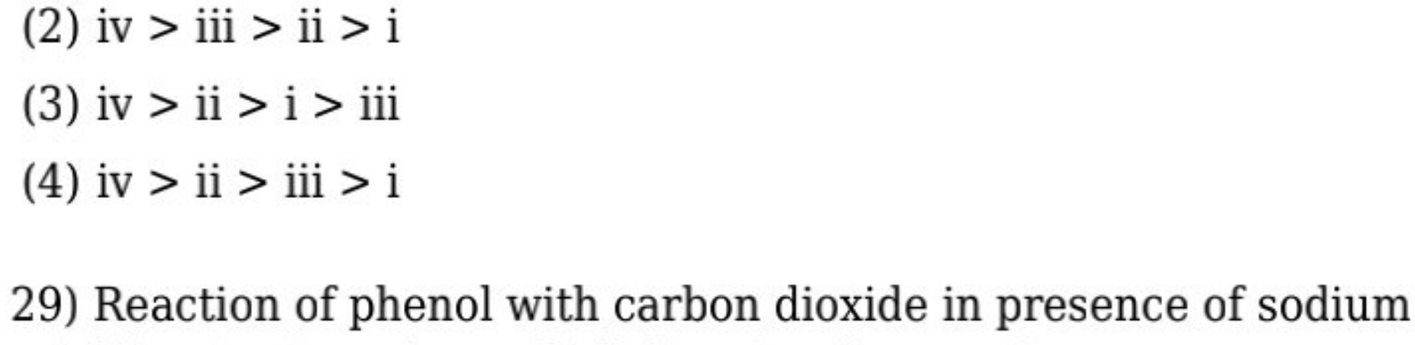
(1) – CHO at ortho position of phenol.

(2) – COOH at ortho position of phenol.

(3) – COOH at meta position of phenol.

(4) – CHO at meta position of phenol.

30) Arrange the following in order of their reactivity when reacting with $HCl/ZnCl_2$:-



I II III IV

(1) IV > I > II > III

(2) I > II > III > IV

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(3) IV > III > II > I

(4) III > II > I > IV

31) **Statement-I :-** Dipole moment of CH_3–Cl is greater than that of CH_3–F.
Statement-II :- The reaction of primary and secondary alcohol with H-Cl require presence of catalyt $ZnCl_2$.

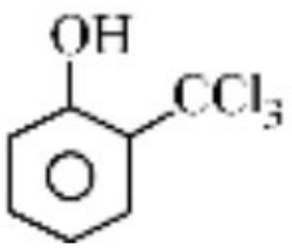
(1) Statement I & II both are correct

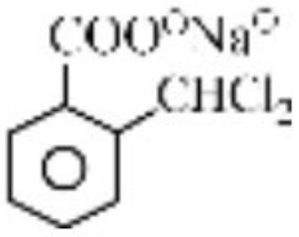
(2) Statement I & II both are incorrect

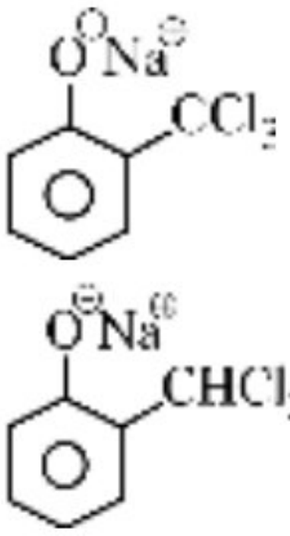
(3) Statement I is correct and II is incorrect

(4) Statement II is correct & I is incorrect

32) Which of the following is intermediate of Reimer Tiemann reaction?

(1) 

(2) 

(3) 

(4)

33) Which of the following reagents is not suitable to oxidise primary alcohols to aldehydes ?

(1) CrO_3 in anhydrous medium

(2) $KMnO_4$ in acidic medium

(3) Pyridinium chlorochromate

(4) Heat in the presence of Cu at 573 K

34) Which of the folowing alkanols is most soluble in water

(1) 1–Butanol

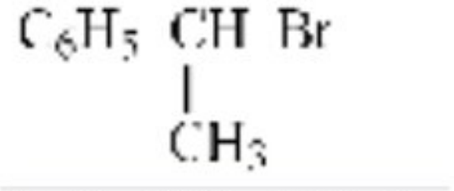
(2) 2–Butanol

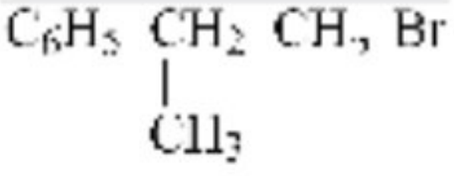
(3) Isobutyl alcohol

(4) t–Butyl alcohol

35) Which of the following are benzylic Halide ?
(A) C_6H_5–CH_2–CH_2Br
(B) C_6H_5–CH_2Br

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(C) 

(D) 

(1) A, B, C, D

(2) B, C

(3) A, D

(4) C, D

36)

Column-I		Column-II	
(A)	n-propyl alcohol and isopropyl alcohol	(P)	Tollen's Test
(B)	Acetaldehyde and acetone	(Q)	Fehling Test
(C)	Acetaldehyde and Benzaldehyde	(R)	Bromine water Test
(D)	Carbolic acid and benzene carboxylic acid	(S)	Lucas test

(1) A → S ; B → Q ; C → P ; D → R

(2) A → R ; B → P, Q ; C → P, Q ; D → S

(3) A → S ; B → P, Q ; C → P, Q ; D → R

(4) A → S ; B → P, Q ; C → Q ; D → R

37) 0.1 gm of organic compound was analysed by Kjeldahl's method. In analysis produced NH_3 absorbed in 20 ml N/5 H_2SO_4. The remaining acid required 20 ml N/10 NaOH for neutralisation. Calculate percentage of nitrogen in organic compound :-

(1) 28 %

(2) 56 %

(3) 35 %

(4) 70 %

38) In the Lassaigne's test for nitrogen in an organic compound, the prussian blue color is obtained due to formation of :-

(1) $Na_4[Fe(CN)_6]$

(2) $Fe_4[Fe(CN)_6]_3$

(3) $Fe_2[Fe(CN)_6]$

(4) $Fe_3[Fe(CN)_6]_4$

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39) Match the column

	Column-I (Mixture)		**Column-II (Purification technique)**
A	Chloroform & aniline	P	Distillation
B	Aniline & water	Q	Chromatography
C	Amino acid	R	Steam distillation
D	Glycerol from spent lye	S	Distillation under reduced pressure

Which of the following is correct.

(1) A-P; B-Q; C-R; D-S

(2) A-P; B-S; C-R; D-Q

(3) A-P; B-R; C-Q; D-S

(4) A-R; B-S; C-P; D-Q

40) Which of following is not correctly match :-

(1) Column chromatography – Al_2O_3 is stationary phase

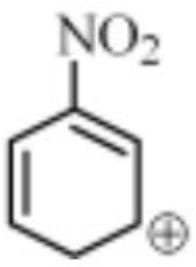
(2) Paper chromatography – Stationary phase is water

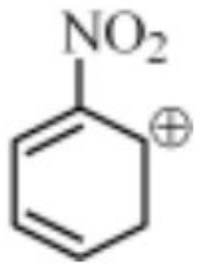
(3) Thin layer chromatography – Example of partition chromatography

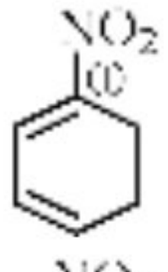
(4) Paper chromatography is example of partition chromatography

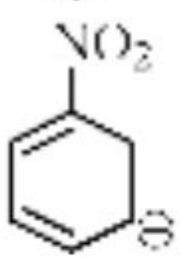
41)

Which is most stable carbocation ?

(1) 

(2) 

(3) 

(4) 

42) Which of the following is an aromatic compound :-

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(1) 

(2) 

(3) 

(4) 

43) Arrange the following compounds in order of their increasing boiling points :-

(a) pentan-1-ol (b) n-butane

(c) pentanal (d) ethoxyethane

(1) a < b < c < d

(2) b < d < c < a

(3) b < a < d < c

(4) b < d < a < c

44) Keto-enol tautomerism is not observed in :-

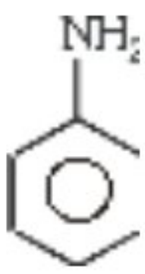
(1) $C_6H_5COC_6H_5$

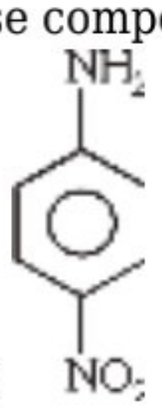
(2) $C_6H_5COCH_3$

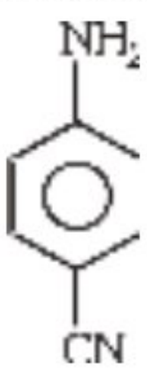
(3) $C_6H_5COCH_2COCH_3$

(4) $CH_3COCH_2COCH_3$

45) Arrange these compounds in decreasing order of basic strength

(i) 

(ii) 

(iii) 

(iv) 

(1) (i) > (ii) > (iii) > (iv)

(2) (ii) > (iii) > (i) > (iv)

(3) (iv) > (i) > (iii) > (ii)

(4) (iv) > (i) > (ii) > (iii)

BIOLOGY

1) A vertebrate which doesn't possess jaws :

(1) *Balanoglossus*

(2) *Petromyzon*

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(3) *Salamandra*

(4) *Scoliodon*

2) Which one is correct about Porifera :-

(1) Unisexual

(2) Internal fertilisation with indirect development

(3) External fertilisation with direct development

(4) External fertilisation with indirect development

3) The characteristic of class Reptilia are :-

(1) Body covered with moist skin which is devoid of scales, the ear is represented by tympanum, alimentary canal, urinary and reproductive tract open into a common cloaca.

(2) Fresh water, animals with bony endoskeleton, air bladder to regulate buoyancy.

(3) Marine animals with cartilagineous endoskeleton.

(4) Body covered with dry and cornified skin, with epidermal scales or scutes

4) Open blood vascular system without any respiratory pigment is found in :-

(1) Earthworm

(2) Cockroach

(3) *Neries*

(4) Hydra

5) Which of the following is **not true** about chondrichthyes ?

(1) Mouth is located ventrally.

(2) Notochord is persistent throughout life.

(3) Teeth are modified placoid scales.

(4) Air bladder is present.

6) Which of the following sentence is incorrect regarding Hemichordata ?

(1) They have open type of circulatory system

(2) Respiration take place through Gills

(3) Fertilization is external in them

(4) Their excretory organ is protonephridia

7) In which animal, the body is divisible into only two body parts that is head and trunk :

(1) *Neophron*

(2) *Pterophyllum*

(3) *Hyla*

(4) *Hemidactylus*

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8) Insects are characterised by presence of three pair of legs. Which of the following animals is an arthropod, but not insect?

(1) *Locusta*

(2) *Bombyx*

(3) *Culex*

(4) *Scorpion*

9) Choanocyte is present in ___A___ and it line __B__ select the option, which correctly fills A and B :-

(1) A- *Spongilla*, B- Bodywall

(2) A - *Ctenoplana*, B - Spongocoel

(3) A - *Sycon*, B - Spongocoel

(4) A-*Hydra*, B-Gastrovascular cavity

10) Animals in which body cavity is not lined by mesoderm, instead, the mesoderm is present as scattered pouches in between the ectoderm and endoderm. Such type of body cavity is called as :-

(1) Schizocoel

(2) Enterocoel

(3) Pseudocoel

(4) Haemocoel

11) Which of the following character is correctly matched with its general character ?

(1) Cyclostomata : cranium and vertebral column are bony

(2) Chondrichthyes : Gill slits without operculum

(3) Amphibia : they shed their scales as their skin casting

(4) Mammalia : Long bones hollow with air cavities

12)

Match column-I with column-II and select the correct option :

Column-I		**Column-II**	
(a)	Phylum	(P)	Agnatha
(b)	Subphylum	(Q)	Cyclostomata
(c)	Division	(R)	Tetrapoda
(d)	Super class	(S)	Chordata
(e)	Class	(T)	Vertebrata

	a	**b**	**c**	**d**	**e**
(1)	S	T	P	Q	R
(2)	S	T	Q	R	P

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(3)	S	T	P	R	Q
(4)	S	R	Q	P	T

(1) 1

(2) 2

(3) 3

(4) 4

13) **Assertion :-** *Scoliodon* has to swim constantly to avoid sinking.
Reason :- In *scoliodon* air bladder is absent.

(1) Both Assertion & Reason are True and the Reason is a correct explanation of the assertion

(2) Both Assertion & Reason are True and the Reason is not a correct explanation of the assertion

(3) Assertion is true but Reason is false

(4) Both Assertion & Reasons are false

14) Read the following statements from A-D ?
(A) They have pneumatic bones
(B) Skin is dry without glands.
(C) Digestive tract has additional chambers, the crop and gizzard.
(D) They are warm blooded.
These statements are correct for which animal ?

(1) *Vipera*

(2) *Columba*

(3) *Labeo*

(4) *Pteropus*

15) Read the following statements.
(a) Metagenesis is observed in Helminths.
(b) Echinoderms are triploblastic and coelomate animals.
(c) Round worms have organ-system level of body organization.
(d) Comb plates present in ctenophores help in digestion.
(e) Water vascular system is characteristic of Echinoderms.
Choose the **correct** answer from the options given below.

(1) (c), (d) and (e) are correct

(2) (a), (b) and (c) are correct

(3) (a), (d) and (e) are correct

(4) (b), (c) and (e) are correct

16) Match the column I, II and III & choose the correct answer :-

	Column-I Animal	**Column-II Feature**	**Column-III Phylum/Class**
(1)	*Psittacula*	Warm blooded	Hemichordata

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(2)	*Myxine*	Nerve cord absent	Cephalochordata
(3)	*Ichthyophis*	Limbs absent	Amphibia
(4)	*Prawn*	Chitinous endoskeleton	Chordata

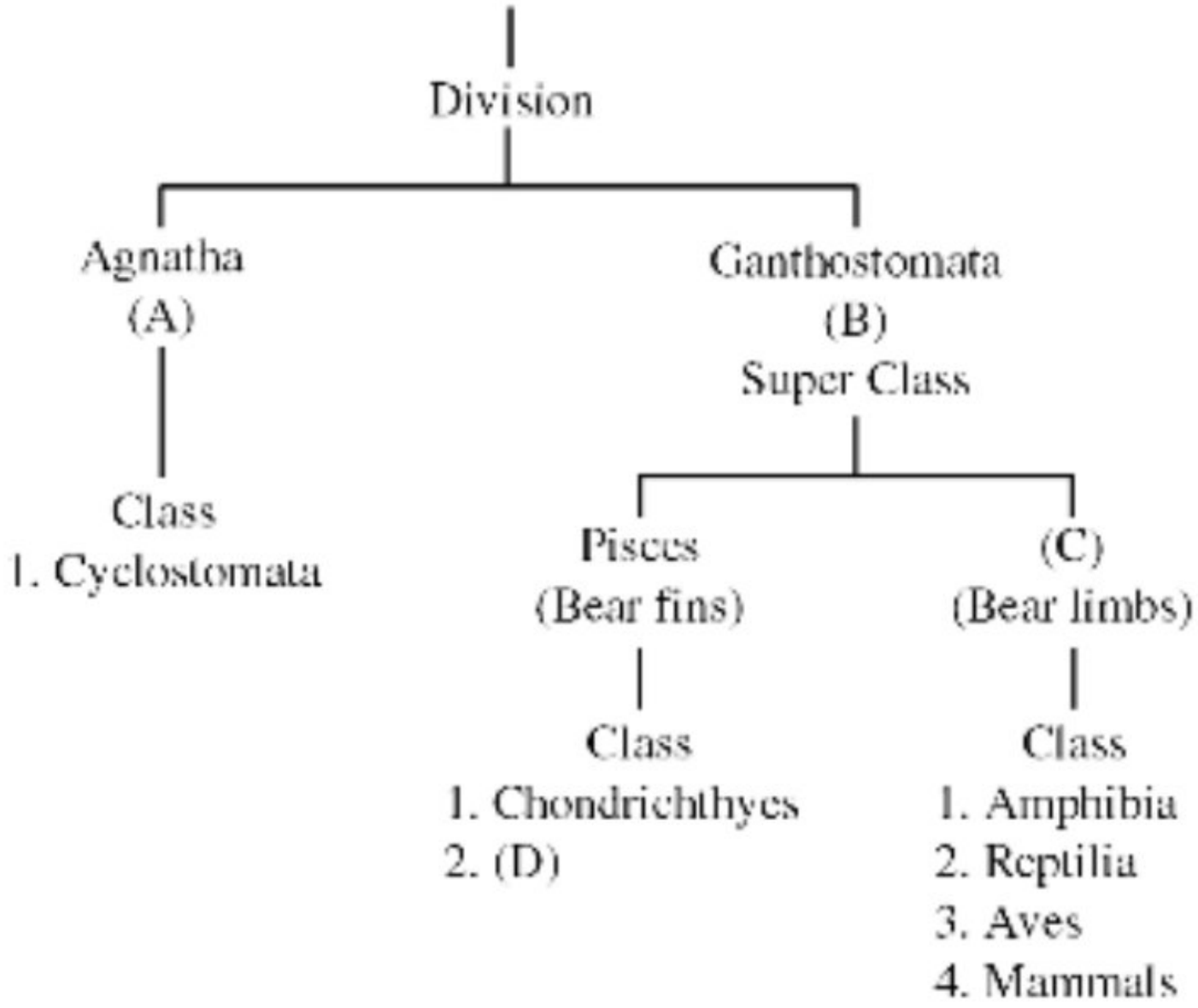
(1) 1

(2) 2

(3) 3

(4) 4

17) Read the flow chart given below and identify A, B, C and D:



Options	**A**	**B**	**C**	**D**
(1)	lacks jaw	Bears jaws	Apoda	Osteichthyes
(2)	Bears jaw	Lacks jaw	Octapoda	Osteichthyes
(3)	Lacks jaw	Lacks jaw	Tetrapoda	Osteichthyes
(4)	Lacks jaw	Bears jaw	Tetrapoda	Osteichthyes

(1) 1

(2) 2

(3) 3

(4) 4

18)

Water transport system of porifera not helps in :

(1) Locomotion

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(2) Respiration

(3) Food transport

(4) Gaseous exchange

19) The unique mammalian characteristics are:

(1) Hairs, pinna and indirect development

(2) Pinna, monocondylic skull and mammary glands

(3) Hairs, tympanic membrane and mammary glands

(4) Hairs, pinna and mammary glands

20) Hemichordates:

(1) Are actually partial chordates

(2) Are now placed as separate phylum under non-chordates

(3) Possess well developed notochord

(4) All of the above

21) Nucleoprotein fibres called chromatin fibres can be observed in :-

(1) Interphase

(2) Metaphase

(3) Anaphase

(4) Late prophase

22) Crossing over occurs between :-

(1) Sister chromatids of homologous chromosomes

(2) Non-sister chromatids of homologous chromosomes.

(3) Non-sister chromatids of non-homologous chromosomes

(4) Sister chromatids of non-homologous chromosomes

23) Meiosis ensures the production of __A__ phase in life cycle of sexually reproducing organisms, where as fertilization restores __B__ phase. A and B are

(1) diploid, haploid respectively

(2) haploid, triploid respectively

(3) diploid, triploid respectively

(4) haploid, diploid respectively

24) The gap between meiosis-I and meiosis-II division is called as :-

(1) Synapsis

(2) Interphase

(3) Interkinesis

(4) G_0 phase

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25) In contrast to mitosis, meiosis occurs only in the :-

(1) Haploid cells
(2) Diploid cells
(3) Pentaploid cells
(4) Triploid cells

26) Which one is the correct sequence of a cell cycle ?

(1) $G_1 \rightarrow$ Karyokinesis $\rightarrow$ cytokinesis $\rightarrow G_2 \rightarrow S$
(2) $S \rightarrow G_2 \rightarrow$ Karyokinesis$\rightarrow$ cytokinesis $\rightarrow$ prophase
(3) $G_1 \rightarrow S \rightarrow G_2 \rightarrow$ Karyokinesis $\rightarrow$ cytokinesis
(4) Cytokinesis $\rightarrow$ Karyokinesis $\rightarrow G_1 \rightarrow S \rightarrow G_2$

27) Which one of the following statement is **incorrect**?

(1) M-phase is the most dramatic period of the cell cycle
(2) Cell division is a progressive process
(3) Prophase follows the cytokinesis
(4) Prophase is marked by the initiation of condensation of chromosomal material

28) Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : The individual chromosomes can no longer be seen and chromatin material tends to collect in a mass at two poles in telophase, last stage of karyokinesis
Reason (R) : The chromosomes that have reached their respective poles decondense and lose their individuality as discrete elements in telophase
In the light of the above statements. Choose the correct answer from the options given below :

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(2) (A) is correct but (R) is not correct.
(3) (A) and (R) both are incorrect.
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

29) **Assertion :-** During G_1 phase the cell is metabolically active and continuously grows but does not replicate its DNA.
Reason :- S or synthesis phase marks the period during which DNA synthesis or replication takes place.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

30) Read the following statements and choose the option which is true for them.
Statement A : The steps of cell division are themselves under genetic control.

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Statement B : Cell growth in terms of cytoplasmic increase is a continuous process.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both the statement are correct

(4) Both the statements are incorrect

31)

Arrange the following events of meiosis in the correct sequence :-
(A) Spliting of the centromere
(B) Terminalisation of the chiasmata
(C) Synapsis
(D) Dyad of the cell
(E) Dissolution of synaptonemal complex
(F) Crossing over

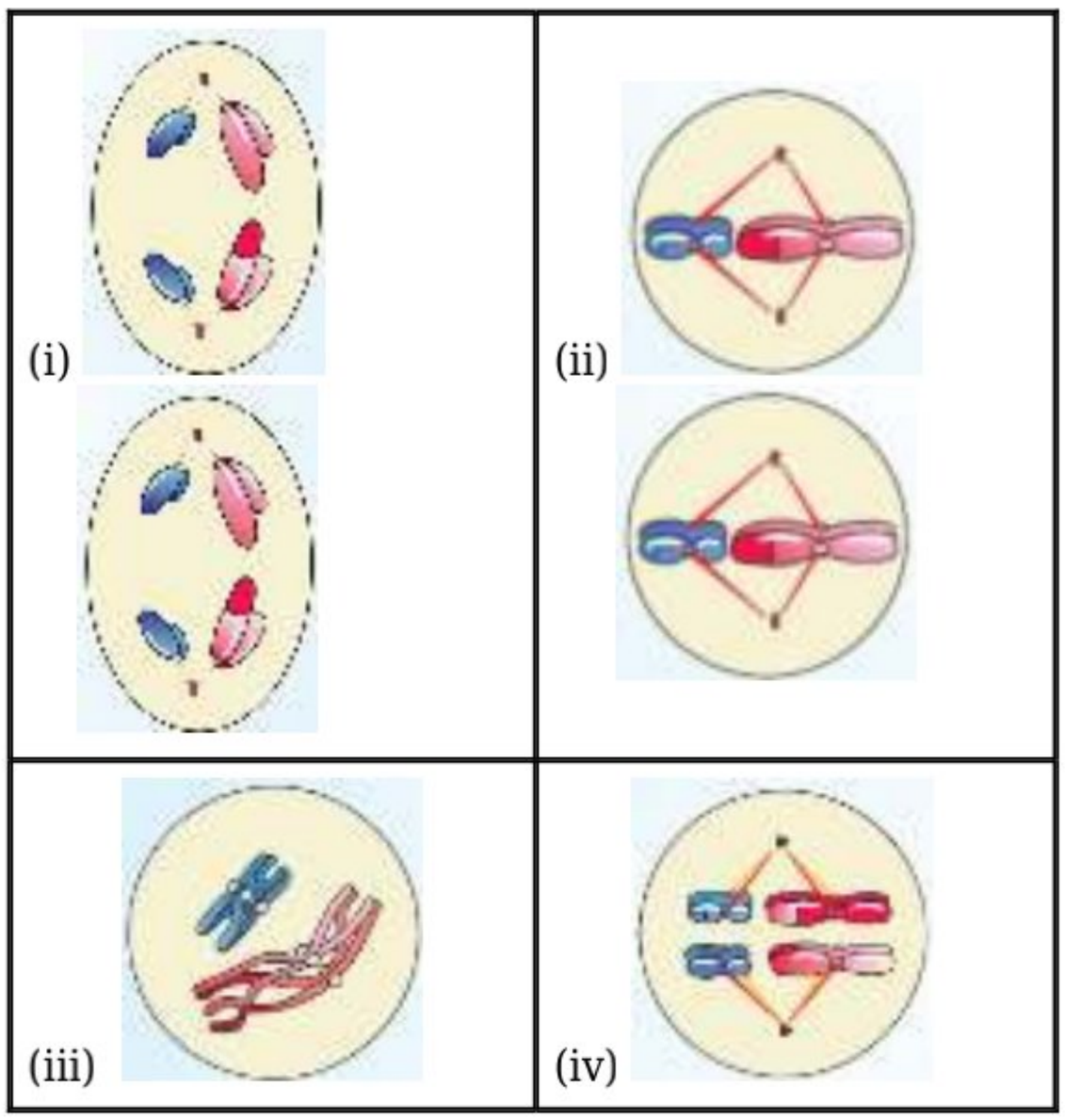
(1) E → C → B → D → A → F

(2) C → E → F → B → D → A

(3) C → F → B → E → A → D

(4) C → F → E → B → D → A

32) Arrange the different stages of meiosis, (i), (ii), (iii) and (iv) in sequence :



(1) (i) → (iv) → (ii) → (iii)

(2) (iii) → (iv) → (ii) → (i)

(3) (iv) → (iii) → (ii) → (i)

(4) (ii) → (i) → (iii) → (iv)

33) Which of the following is not associated with G. N. Ramachandran?

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(1) He gave triple helical structure of collagen protein

(2) He gave Ramachandran plot for analysis of allowed conformation of protein

(3) Founder of Madras school of conformational analysis of biopolymers

(4) He gave concept of reductionist biology

34) Cell membrane in prokaryotic and eukaryotic cell respectively are:

(1) Permeable and impermeable

(2) Selectively permeable and impermeable

(3) Selectively permeable in both

(4) Semi permeable and impermeable

35) Extensive compartmentalization in eukaryotic cell occurs due to presence of:

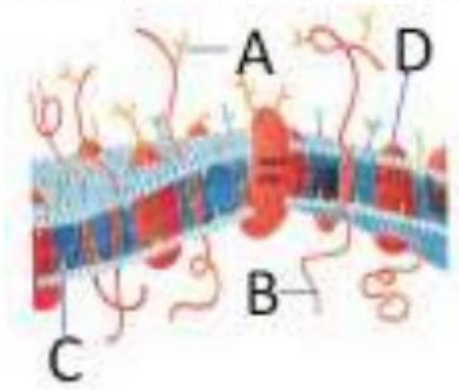
(1) Single membrane bound cell organelle only

(2) Membrane less cell organelle

(3) Membrane bound cell organelle

(4) Double membrane bound cell organelle only

36) Which of the following structure in given diagram provide membrane stability?



(1) A

(2) B

(3) C

(4) D

37) Which of the member of endomembrane system is associated with protein synthesis?

(1) Ribosomes

(2) RER

(3) SER

(4) Golgi body

38) Major site of glycoprotein synthesis in plant cell is:

(1) RER

(2) Golgi body

(3) Vacuole

(4) Ribosome

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39) Which of the following is correctly matched?

(1) Sap vacuole: Storage of waste material in animal cell
(2) Contractile vacuole: Osmoregulation
(3) Gas vacuole: Storage of gas and food
(4) Food vacuole: Present in BGA

40) Space limited by inner membrane of chloroplast is known as:

(1) Protoplast
(2) Matrix
(3) Tonoplasm
(4) Stroma

41) Which of the following was discovered by George Pallade using electron microscope?

(1) Centrosome
(2) Ribosome
(3) Lysosome
(4) ER

42) Cilia and flagella are hair like outgrowth of cell membrane and mainly made-up of microtubules. What are the microtubular arrangement in structure which give rise to cilia and flagella?

(1) 9 Doublet + 2 Singlet
(2) 9 Triplet+0
(3) 9 triplet + 2 Singlet
(4) 9 Doublet +0

43) Match the column I with column II

Column I		**Column II**	
A	Living cell	P	Flemming
B	Nucleus	Q	Leeuwenhoek
C	Chromatin	R	Rudolf Virchow
D	Final shape to cell theory	S	Robert Brown

(1) A: Q; B: S; C: P; D: R
(2) A: Q; B: S; C: R; D: P
(3) A: S; B: Q; C: P; D: R
(4) A: Q; B: S; C: R; D: P

44) **Assertion (A):** Cell wall protects the plant cell from mechanical injury and infections.
Reason (R): Cell wall provide and maintain shape to plant cell.

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(1) A and R both are true and R explain A

(2) A is true but R is false

(3) A is false but R is true

(4) A and R both are true but R does not explain A

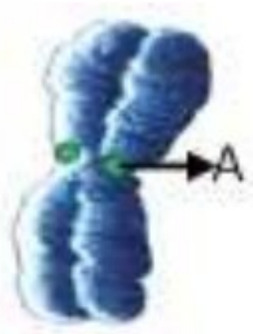
45) A structure/cell when viewed under microscope does not shows nucleus but show the cellulosic cell wall and central vacuole. This cell/structure may be:

(1) Mature RBC of mammals

(2) Blue Green Algae cell

(3) Mature sieve tube element of angiosperm

(4) Either blue green algae or mature sieve tube element



46) The structure A labelled in given diagram is site for:

(1) Attachment of spindle fibre

(2) Attachment of chromatid

(3) Attachment of centriole

(4) Attachment of astral ray

47) During meiosis simultaneous splitting of chromosomes occurs in:

(1) Anaphase II

(2) S Phase

(3) Anaphase I

(4) Telophase

48) 23 chromosomes are present in sperm cell of human. How many chromosomes and chromatids are respectively present in human during metaphase I stage?

(1) 23 and 46

(2) 46 and 46

(3) 46 and 92

(4) 92 and 92

49) Which of the following occurs in S phase of cell cycle in cytoplasm of an animal cell?

(1) DNA replication

(2) Centriole duplication

(3) Condensation of chromosomes

(4) 1 and 2 both

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50) Although chromosome may undergo some dispersion but it will not reach up to extremely extended state and dyad of cells are formed in:

(1) Telophase II

(2) Telophase of mitosis

(3) Telophase I

(4) Anaphase I

51) Consider the followings :
Solanum tuberosum, Mangifera indica, Triticum aestivum, Solanum nigrum, Panthera leo, Panthera pardus, Solanum melongana, Panthera tigris, Homo sapiens, Musca domestica
How many family, genus and species are represented by above set of organisms ?

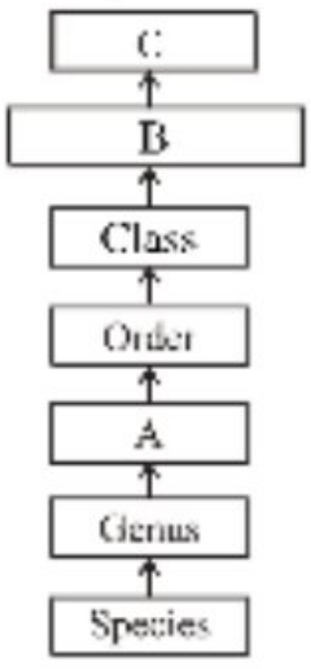
(1) 5, 5, 9 respectively

(2) 6, 6, 10 respectively

(3) 10, 6, 5 respectively

(4) 6, 6, 9 respectively

52) The given flowchart represents the hierarchy of various taxonomic categories.
Identify the missing categories (A,B and C) and select the correct statements regarding these.



(i) A is the taxonomic category which contains a number of related genera.
(ii) Examples of category B are Monocotyledoneae, Dicotyledoneae, Arthropoda, etc.
(iii) C represents the largest unit of taxonomic hierarchy.
(iv) Examples of category C are Fungi, Algae, Bryophyta etc.

(1) (i) and (ii)

(2) (iii) and (iv)

(3) (i) and (iii)

(4) (i), (ii), (iii) and (iv)

53) Consider the following statements (A-D) regarding biological names :-
(A) Both the words in a biological name, when typed or handritten are separately underlined.
(B) Biological names are made up of two letters.
(C) The first word denoting the Genus starts with a capital letter while specific epithet starts with a small letter.
(D) Binomial nomenclature system was given by Carolus Linnaeus.
Which of the above statement(s) is/are correct?

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(1) Only A & B

(2) Only B, C & D

(3) Only A, C & D

(4) A, B, C & D

54) In which of the following gymnosperms, corolloid root having N_2-fixing cyanobacteria (*Nostoc*) are found ?

(1) *Pinus*

(2) *Ginkgo*

(3) *Cycas*

(4) *Cedrus*

55) Biodiversity refers to :-

(1) Only number of organisms present on earth

(2) Only types of organisms present on earth

(3) Number and types of organisms present on earth

(4) Number and types of dead organisms present in rivers

56) Hominidae is placed between :-

(1) Genus and Species

(2) Order and Class

(3) Class and Family

(4) Order and Genus

57) Two different organisms belong to same class but not the same family, then these two organisms may belong to the same.

(1) Genus

(2) Order

(3) Species

(4) Sub-species

58) Select the mismatch pair w.r.t organism and its order :

(1) Wheat – Poales

(2) Mango – Sapindales

(3) Housefly – Diptera

(4) Man – Mammalia

59) In two-kingdom classification *Penicillium* was placed in :

(1) Kingdom Plantae

(2) Kingdom Animalia

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(3) Kingdom Protista

(4) Kingdom fungi

60) In which kingdom would you classify the Archaebacteria and nitrogen-fixing organisms, if the five-kingdom system of classification is used :-

(1) Plantae

(2) Fungi

(3) Protista

(4) Monera

61) Bacterial mesosome is made up of :-

(1) Amino acid + polysaccharide

(2) Lipid + protein

(3) Lipid + polysaccharide

(4) Phospholipid only

62)

Choose the correct match for bacteria :-

(1) Pili – Help in locomotion

(2) Glycocalyx – Help in genetic recombination

(3) Cell wall – Help to prevent from bursting or collapsing

(4) Plasmid – Help in colony formation

63) Which of the following have capacity of N_2 fixation?

(1) *Anabaena*

(2) *Rhizobium*

(3) *Nostoc*

(4) All of these

64) The spores of slime moulds are dispersed by :

(1) Air

(2) Water

(3) Insects

(4) Birds

65) Ecological role of bacteria?

(1) Curd making

(2) Antibiotic production

(3) Flavouring of tea

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(4) Recycling nutrients

66) According to five kingdom system single celled eukaryotes are placed under Protista, members of Protista are primarily :-

(1) Terrestrial
(2) Aquatic
(3) Saprophytic
(4) Parasitic

67) Basidiospores and ascospores are produced :-

(1) Exogenously, Endogenously respectively
(2) Exogenously, Exogenously respectively
(3) Endogenously, Endogenously respectively
(4) Endogenously, Exogenously respectively

68) Disease causing eukaryotes are include under :

(1) Fungi
(2) Virus
(3) Bacteria
(4) Prions

69) Mad cow disease in catttle is caused by an organism which has :-

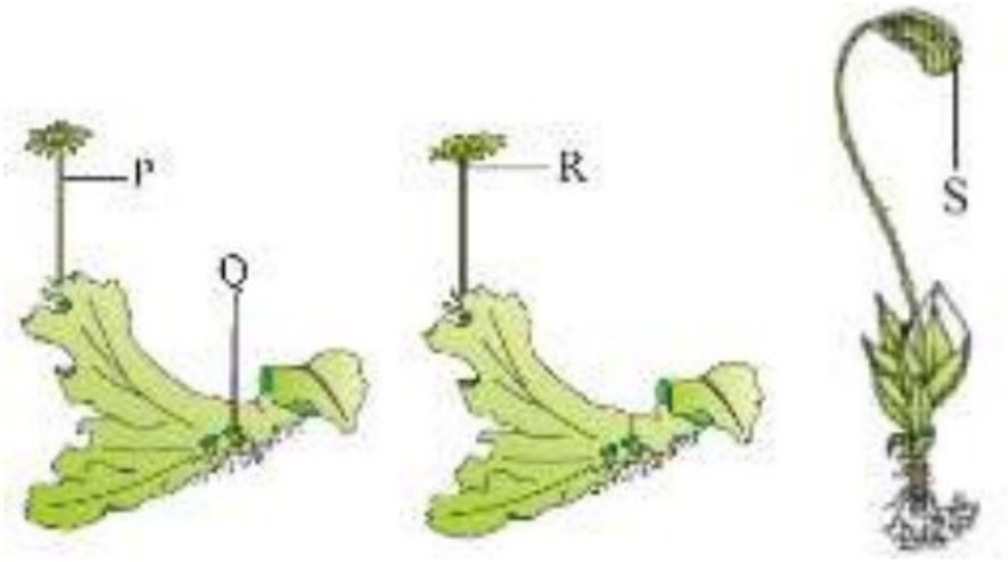
(1) Inert crystalline structure
(2) Abnormally folded protein
(3) Free RNA without protein coat
(4) Free DNA without protein coat

70) Which of the following is present in gut of several ruminants ?

(1) *Paramoecium*
(2) Slime moulds
(3) Methanogens
(4) Cyanobacteria

71) Observe the following figures and identify the labelled structures P, Q, R and S :-

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	P	Q	R	S
(1)	Archegonio- phore	Gemma cup	Antheridio- phore	Seta
(2)	Antheridio-phores	Rhizoids	Archegonio-phore	Capsule
(3)	Antheridio-phore	Seta	Archegonio-phore	Leaves
(4)	Archegonio-phore	Gemma cup	Antheridio-phore	Capsule

(1) 1

(2) 2

(3) 3

(4) 4

72) Which of the following is a flagellated protozoan ?

(1) *Treponema*

(2) *Trypanosoma*

(3) *Trichoderma*

(4) *Entamoeba*

73) Agar-agar, a complex carbohydrate, is extracted from :

(1) Blue green algae - *Nostoc*

(2) Red Algae - *Polysiphonia*

(3) Red Algae - *Gelidium*

(4) Green Algae - *Chlorella*

74) Identify the following figures A and B respectively :-

(A) (B)



(1) A - *Euglena,* B - *Aspergillus*

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(2) A - *Paramoecium*, B - *Mucor*

(3) A - *Paramoecium*, B - *Agaricus*

(4) A - *Euglena*, B - *Agaricus*



75) Answer the following questions (a, b and c) by using the given diagram :- (a) Name of plant ?

(b) Ploidy of plant ?

(c) Plant is homosporous or heterosporous ?

	(a)	(b)	(c)
(1)	*Selaginella*	Diploid	Heterosporous
(2)	*Salvinia*	Diploid	Heterosporous
(3)	*Pinus*	Haploid	Homosporous
(4)	*Salvinia*	Haploid	Homosporous

(1) 1

(2) 2

(3) 3

(4) 4

76) Basidiocarp and ascocarp are the fruiting bodies of fungi. They can be seen in :-

(1) *Alternaria* & *Agaricus* respectively

(2) *Alternaria* & *Ustilago* respectively

(3) *Claviceps* & *Agaricus* respectively

(4) *Agaricus* & *Claviceps* respectively

77) Mycoplasma is :-

(1) Eukaryotic cell with cell wall

(2) Prokaryotic cell with cell wall

(3) Eukaryotic cell without cell wall

(4) Prokaryotic cell without cell wall

78) Which of the features of *Anabaena* separate it from *Agaricus*.

(1) Presence of cell membrane

(2) Presence of cell wall

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(3) Presence of chlorophyll a

(4) Presence of cytoplasm

79) Which of the following fungi is mainly responsible for mineral cycling -

(1) *Puccinia*

(2) *Alternaria*

(3) *Ustilago*

(4) Morels

80) **Statement-1 :** In Ascomycetes and Basidiomycetes, an intervening dikaryophase occurs.
Statement-2 : Dikaryotic mycelium ultimately gives rise to basidium, in Basidiomycetes.

(1) Both the statement are correct.

(2) Both the statement are incorrect.

(3) Statement-1 is correct but Statement-2 is incorrect.

(4) Statement-1 is incorrect but Statement-2 is correct.

81) **Statement-I :** The morphology of mycelium form the basis of the classification of the Kingdom Fungi.
Statement-II : Mode of spore formation form the basis of the classification of the Kingdom Fungi.

(1) Both Statement-I and Statement-II are true.

(2) Both Statement-I and Statement-II are false.

(3) Statement-I is true but Statement-II is false.

(4) Statement-I is false but Statement-II is true.

82) **Assertion (A) :-** Cell wall of Diatoms are indestructible.
Reason (R) :- Cell wall of Diatoms have layer of calcium pectate embedded in it.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) Both (A) and (R) are false.

83) Identify the plant as showing in the figure and select the incorrect statement regarding this :–



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(1) They lack true root, stem, or leaves.

(2) Produce biflagellate antherozoids.

(3) In this, Zygotes undergo reduction division immediately.

(4) They play an important role in plant succession on bare rocks/soil.

84) Select the incorrect statement with respect to bryophytes :-

(1) Thallus is attached to the substratum by rhizoids.

(2) The sporophyte is free-living and independent.

(3) Sex organs are multicellular.

(4) Spores germinate to produce gametophyte.

85)

Which of the following statment is not correct ?

(1) Diatoms are chief producers of ocean.

(2) *Selaginella* is a heterosporous fern.

(3) Spores of slime moulds are not devoid of cell wall.

(4) The stems of *Pinus* are branched.

86) Read the following statements and select incorrect only :-
(a) About 70 percent species of marine algae are used as food.
(b) A few algae of marine forms such as kelps, form massive plant body.
(c) *Eudorina* shows anisogamous type of sexual reproduction.
(d) In red algae the reserve food is mannitol.
Option :-

(1) c and d

(2) a and d

(3) a and b

(4) b and c

87) **Assertion (A) :** In green algae vegatative reproduction usually takes place by fragmentation.
Reason (R) : In green algae asexual reproduction occurs by flagellated zoospores.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

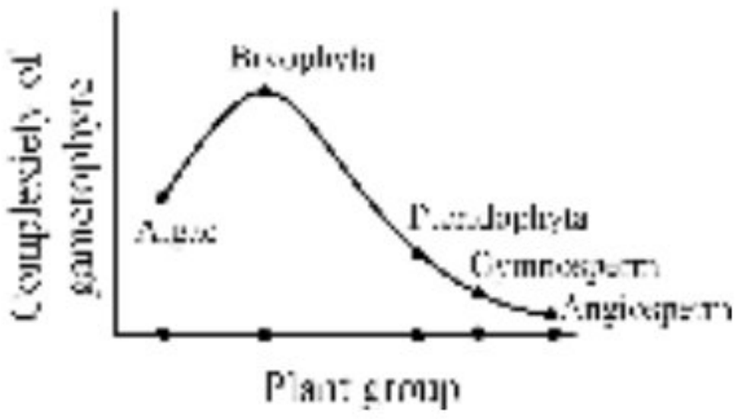
(3) (A) is true but (R) is false.

(4) Both (A) and (R) are false.

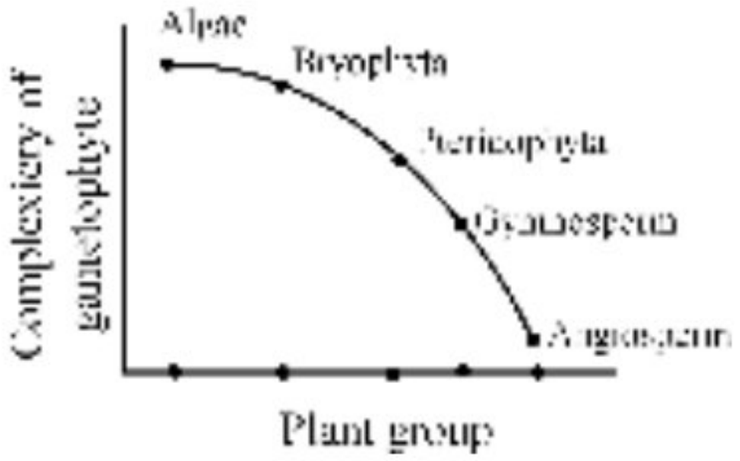
88) Which of the following graph correctly represents the evolution of gametophytic generation in different plant groups ?

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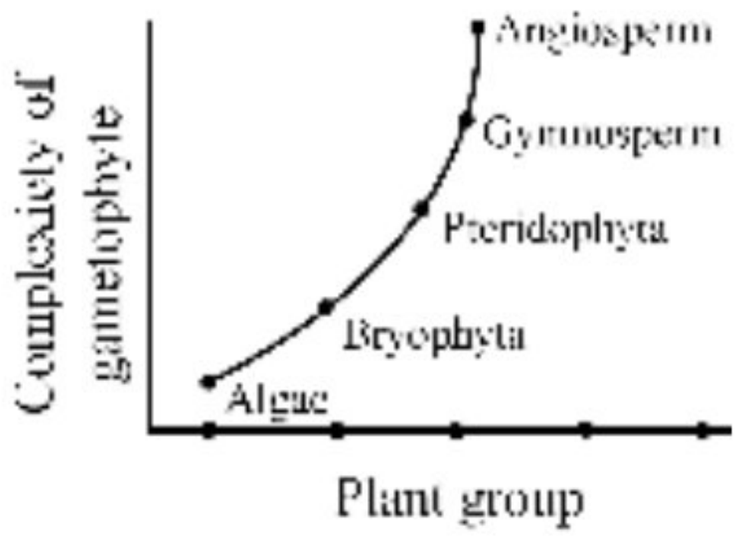
(1)



(2)



(3)



(4)



89) Match **List-I** with **List-II**

List-I		**List-II**	
A.	Pteridophyte	I.	*Spirogyra*
B.	Bryophyte	II.	*Ginkgo*
C.	Algae	III.	*Polytrichum*
D.	Gymnosperm	IV.	*Salvinia*

Choose the option with all **correct** matches :

(1) A-III, B-IV, C-II, D-I

(2) A-IV, B-III, C-I, D-II

(3) A-III, B-IV, C-I, D-II

(4) A-IV, B-III, C-II, D-I

90)

Match List-I with List-II :

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	List-I		List-II
(A)	Ascomycetes	(I)	*Albugo*
(B)	Basidiomycetes	(II)	*Neurospora*
(C)	Deuteromycetes	(III)	*Ustilago*
(D)	Phycomycetes	(IV)	*Trichoderma*

Choose the option with all correct matches :

(1) A-I, B-II, C-IV, D-III

(2) A-II, B-III, C-IV, D-I

(3) A-I, B-II, C-III, D-IV

(4) A-II, B-III, C-I, D-IV

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ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	**2**	**4**	**4**	**2**	**1**	**3**	**2**	**4**	**1**	**3**	**1**	**2**	**2**	**1**	**4**	**3**	**2**	**2**	**4**	**1**
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	**2**	**3**	**4**	**1**	**2**	**4**	**1**	**2**	**3**	**4**	**2**	**1**	**1**	**1**	**3**	**4**	**4**	**1**	**2**	**2**
Q.	41	42	43	44	45															
A.	**4**	**1**	**4**	**1**	**4**															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	**3**	**1**	**1**	**4**	**2**	**3**	**3**	**3**	**1**	**1**	**3**	**3**	**3**	**4**	**4**	**3**	**1**	**4**	**1**	**3**
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	**3**	**2**	**2**	**2**	**3**	**3**	**2**	**4**	**2**	**1**	**1**	**4**	**2**	**4**	**2**	**4**	**1**	**2**	**3**	**3**
Q.	86	87	88	89	90															
A.	**2**	**2**	**2**	**1**	**3**															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	**2**	**2**	**4**	**2**	**4**	**4**	**3**	**4**	**3**	**3**	**2**	**3**	**1**	**2**	**4**	**3**	**4**	**1**	**4**	**2**
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	**1**	**2**	**4**	**3**	**2**	**3**	**3**	**4**	**2**	**3**	**4**	**2**	**4**	**3**	**3**	**3**	**2**	**2**	**2**	**4**
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	**2**	**2**	**1**	**4**	**3**	**1**	**1**	**3**	**2**	**3**	**2**	**3**	**3**	**3**	**3**	**4**	**2**	**4**	**1**	**4**
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	**2**	**3**	**4**	**1**	**4**	**2**	**1**	**1**	**2**	**3**	**4**	**2**	**3**	**3**	**2**	**4**	**4**	**3**	**2**	**1**
Q.	171	172	173	174	175	176	177	178	179	180										
A.	**1**	**3**	**3**	**2**	**2**	**2**	**2**	**1**	**2**	**2**										

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SOLUTIONS

PHYSICS

1) $Q = \int i\,dt$ = Area = $\frac{1}{2}(4)(0.1)$ = 0.2 C

$Q = \frac{\Delta\phi}{R}$

$\Delta\phi$ = QR = (0.2) (10) = 2 Wb

2)

$V = \frac{LdI}{dt}$

3) $\phi = 4t^2 - 4t + 2$

$\frac{d\phi}{dt} = 8t - 4$

t = 1 sec = 8 – 4 = 4 volt

4)

Lenz law : Direction of induced current is such that it oppose increase or decrease the magnetic field in any direction.

5) Acceleration of magnet is less than g because due to induced current in coil magnet will experience an upward force.

6) Resultant voltage = 200 volt
Since V_1 and V_3 are 180° out of phase, the resultant voltage is equal to V_2
$\therefore V_2$ = 200 volt

7) Induced e.m.f. = Blv = $0.3 \times 10^{-4} \times 10 \times 5$
= 1.5×10^{-3} V = 1.5 mV

8) $V = \frac{Ldi}{dt} \Rightarrow 2t = 2 \times \frac{di}{dt}$

$C + \frac{t^2}{2} - i \Rightarrow i = \frac{t^2}{2}$

(1) is correct

$U = \frac{1}{2}Li^2 = \frac{1}{2} \times 2 \times \frac{t^4}{4}$ = 4 joule

(2) is correct

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$$\frac{d(i)}{dt} \quad L.i\ \frac{di}{dt} \quad 2 \times \frac{t^2}{2} \times t \quad 1\ j/s$$

(3) is correct

9)

Since induce current will be in anticlock wise direction.

10) $\frac{V_1}{V_2} = \frac{N_1}{N_2}$

$\frac{230}{V_2} = \frac{10}{1}$

$V_2 = 23\ V$

Power consumed $= \frac{V_2^2}{R}$

$= \frac{23 \times 23}{46} = 11.5\ W$

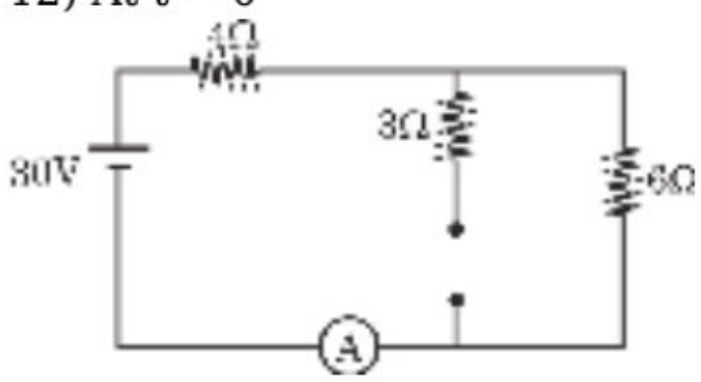
11)

$\phi = Mi = BA$

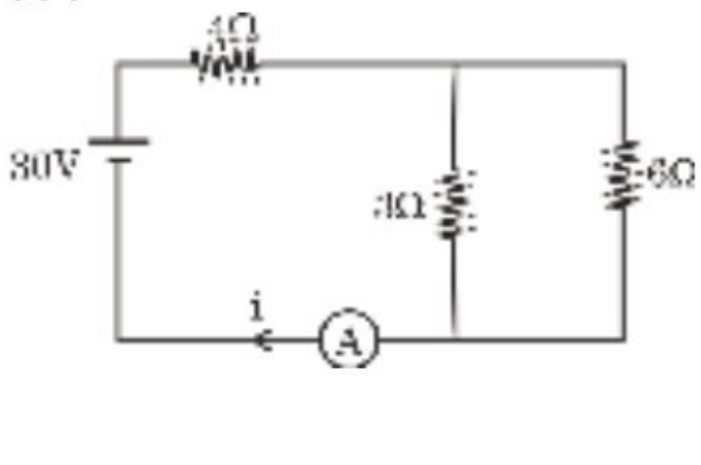
$\Rightarrow Mi = \frac{\mu_0 i}{2b}\pi a^2$

$\therefore M = \frac{\mu_0 \pi a^2}{2b}$

12) At t = 0



at t = ∞



13)

Question Explanation: A square wave current alternates between +0.5 A and –0.5 A; find the AC ammeter reading (RMS value).

Concept: RMS value of alternating current.

Solution: For a square wave, current magnitude is constant.

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So, $I_{rms} = \sqrt{[(1/T) \times \int_0^T i^2(t)\, dt]}$
Since $i = \pm 0.5$ A, $i^2 = (0.5)^2 = 0.25$ A^2 always.
Hence, $I_{rms} = \sqrt{(0.25)} = 0.5$ A

Final Answer: $I_{rms} = 0.5$ A

14) $P_{avg} = V_{rms}\, I_{rms} \cos\theta$

$$= \left(\frac{V_0}{\sqrt{2}}\right)\left(\frac{I_0}{\sqrt{2}}\right)\cos\theta$$

$$= \left(\frac{100}{\sqrt{2}}\right)\left(\frac{20}{\sqrt{2}}\right)\cos 45$$

$$= \frac{1000}{\sqrt{2}} \text{ watt}$$

wattless current = $I_{rms}\ \sin\theta$

$$= \frac{I_0}{\sqrt{2}}\sin\theta$$

$$= \frac{20}{\sqrt{2}}\sin 45°$$

= 10 amp.

15) At $f = 0$, $X_L = 0$, $X_C = \infty$, $\Rightarrow Z = \infty$ ∴ $I = 0$
at $f = \infty$, $X_L = \infty$, $X_C = 0$, $\Rightarrow Z = \infty$, $I = 0$

16) $$V_{rms} = \sqrt{\frac{V_0^2 + 0}{2}} = \frac{V_0}{\sqrt{2}}$$

17) $$z = \sqrt{R^2 + x_L^2}$$

$$z = \sqrt{(100)^2 + (100)^2}$$

$$= \sqrt{2}\,(100)$$

$$\cos\phi = \frac{R}{2} = \frac{100}{100\sqrt{2}} = \frac{1}{\sqrt{2}}$$

18)

From graph

$$E = E_0 \sin\omega t$$

$$i = -i_0 \cos\omega t = i_0 \sin(\omega t + \pi + \pi/2)$$

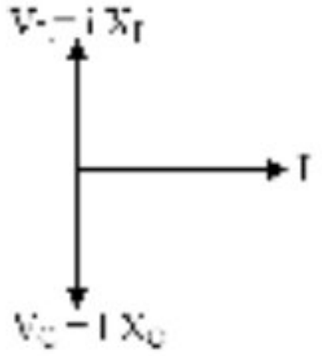
$$= i_0 \sin\left(\omega t + \frac{3\pi}{2}\right)$$

$$i = i_0 \sin\left(\omega t - \frac{\pi}{2}\right)$$

∴ Phase of voltage leads current by π/2

19) Current is same through L & C.
So phases diagram;

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20) using KCL, $I_3 = I_1 + I_2$

$I_3 = 3A \sin \omega t + 4A \cos \omega t$

$I_3 = R \sin (\omega t + \theta)$

where $R = \sqrt{(3A)^2 + (4A)^2} = 5A \quad \tan \theta = \frac{4}{3} \Rightarrow \theta = 53^\circ$

21) Direction of propagation of EM wave will be in the direction of $\vec{E} \times \vec{B}$.

22) speed $= \frac{\text{coefficient of t}}{\text{coefficient of x}}$

$3 \times 10^8 = \frac{6 \times 10^8}{k} \Rightarrow k = 2$

23)

Amplitude of electric field

$E_0 = B_0 C = 60$ V/m

direction of propagation $-\hat{i} = \hat{E} \times \hat{B}$

$\Rightarrow -\hat{i} = \hat{E} \times \hat{k}$

24)

Intensity $= \frac{1}{2} \varepsilon_0 E_0^2 c$

$= \frac{1}{2} \times 9 \times 10^{-12} \times (600)^2 \times 3 \times 10^8$

$= \frac{9}{2} \times 36 \times 3 = 486$ W/m^2

25)

Infrared is the least energetic thus having biggest wavelength (λ) & gamma rays are most energetic thus having smallest wavelength (λ).

26) $\frac{\varepsilon_m \times \mu_m}{\varepsilon_0 \times \mu_0} = \frac{\frac{1}{v^2}}{\frac{1}{c^2}}$

$\varepsilon_r \times \mu_r = \frac{c^2}{v^2}$

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$$\varepsilon_r \times 2 = \frac{(3 \times 10^8)^2}{(1.5 \times 10^8)^2}$$

$$\varepsilon_r \times 2 = 4$$

$$\varepsilon_r = 2$$

27) $f = 3 \times 10^{10}\ H_z$ $\quad B_0 = 10^{-7} T\ E_0 = CB_0 = 3 \times 10^8 \times 10^{-7} = 30\ V/m$

$$\lambda = \frac{C}{f} = 10^{-2} m$$

$E_z = E_0 \sin(Kx - \omega t)$

$$= 30 \sin\left(\frac{2\pi}{\lambda}x - \frac{2\pi}{T}t\right)$$

$$\boxed{E_z = 30 \sin 2\pi(100x - 3 \times 10^{10}t)}\ V/m$$

28) Copper (Cu) is a **metal**, which has a positive temperature coefficient of resistance. This means its resistance decreases as its temperature decreases. Germanium (Ge) is a **semiconductor**, which has a negative temperature coefficient of resistance. Its resistance increases as its temperature decreases because fewer charge carriers are available for conduction.

29) **Case–I :**

$V_A < V_B$ $\quad [\because (-10V) < (-5V)]$

D_1 & D_3 are RB and D_2 is FB

$$\therefore R_1 = \frac{R}{4} + R + \frac{R}{4} = \frac{3R}{2}$$

Case–II :

$V_A > V_B$

$\Rightarrow D_1$ & D_2 are FB and D_3 is RB

$$\therefore R_2 = \frac{R}{4} + \frac{\frac{R}{2} \times \frac{R}{2}}{\left(\frac{R}{2} + \frac{R}{2}\right)} + \frac{R}{4} = \frac{3R}{4}$$

Case–III :

$V_A > V_B$

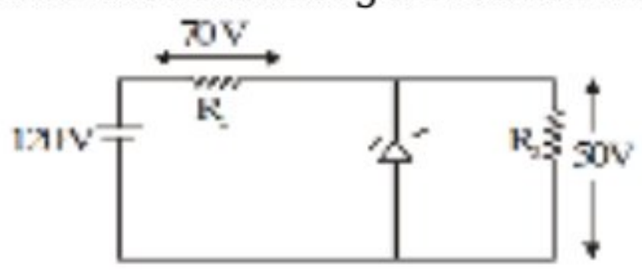
$\Rightarrow$ same as case–II

$R_3 = R$

30) Assuming zener diode doesnot undergo breakdown, current in circuit $= \frac{120}{15000} = 8\,mA$

$\therefore$ Voltage drop across diode = 80 V > 50 V.

The diode undergo breakdown.



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Current is $R_1 = \frac{70}{5000} = 14\text{mA}$

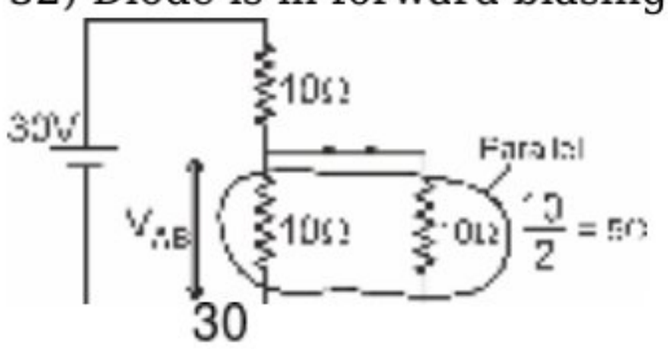
Current is $R_2 = \frac{50}{10000} = 5\,\text{mA}$

 Current through diode = 9mA

31)

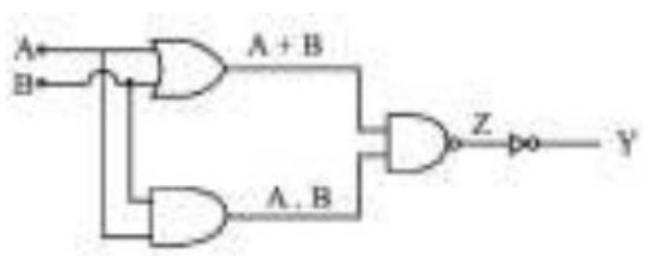
The circuit is a half-wave rectifier.
The diode is in series with the load resistor R (V_{CD} across R).
The output waveform consists only of the **positive half-cycles**.

32) Diode is in forward biasing hence the circuit can be redrawn as follows



$$V_{AB} = \frac{30}{(10+5)} \times 5 = 10\ \text{V}$$

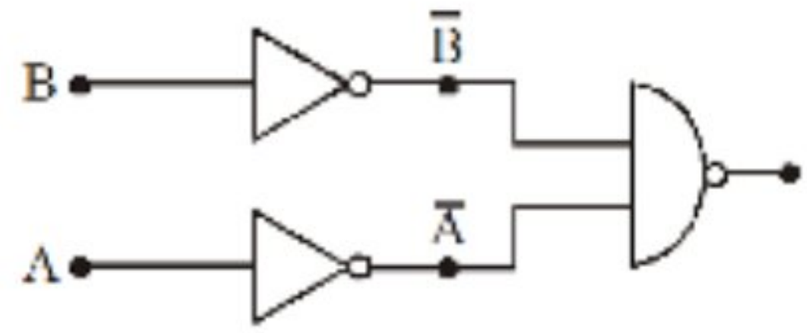
33)



$\overline{Z} = (A + B) . (A . B)$
$Y = \overline{\overline{Z}} = (A + B) . (A . B)$
Y = A. B
 It is an AND Gate

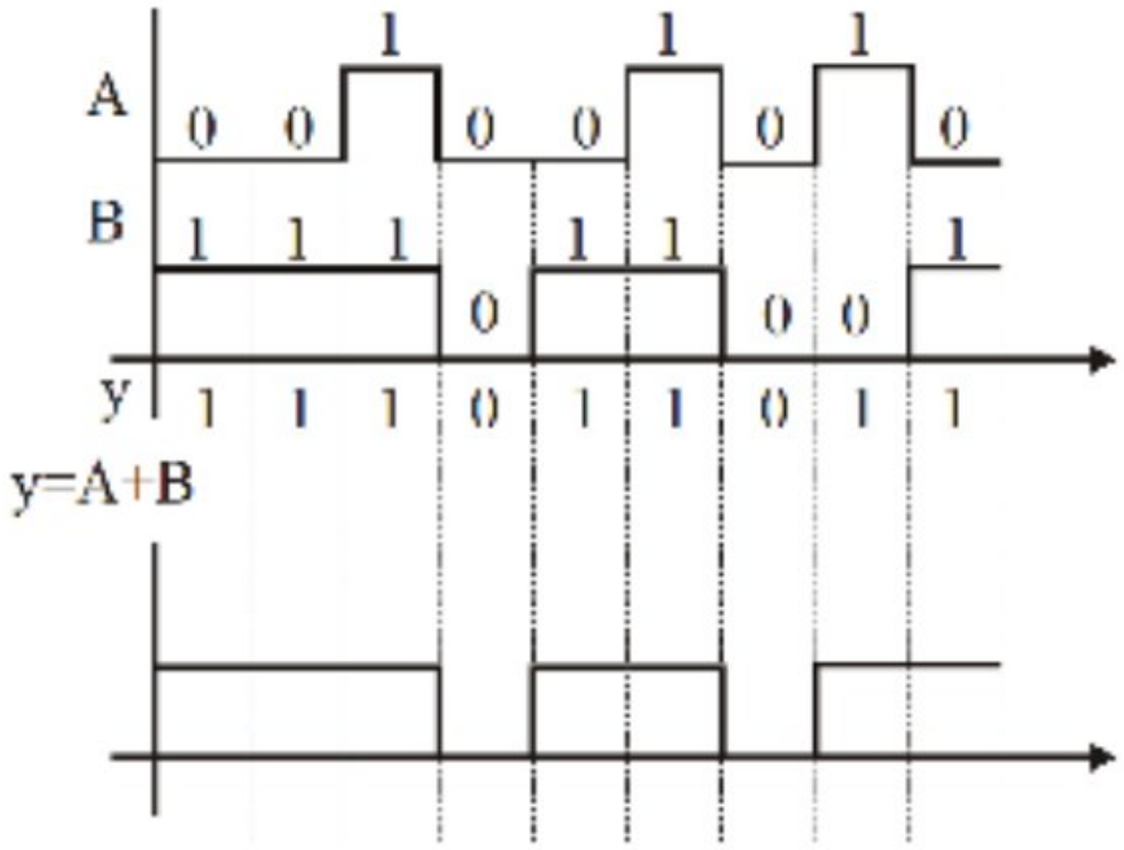
34) We know that for a doped semiconductor in thermal equilibrium, we have
$n_e n_h = n_i^2$
As per given data,
$n_i = 1.5 \times 10^{16}\ m^{-3}$
$n_h = 4.5 \times 10^{22}\ m^{-3}$

Thus $n_e = \frac{n_i^2}{n_h} = \frac{(1.5 \times 10^{16})^2 m^{-6}}{4.5 \times 10^{22} m^{-3}} = 5 \times 10^9 m^{-3}$



35) $y = \overline{\overline{A}.\overline{B}} = \overline{\overline{A}} + \overline{\overline{B}} = A + B$

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36) $\frac{hc}{\lambda} = \phi + (k_{max})_1$

$\frac{4hc}{3\lambda} = \phi + (k_{max})_2$

$\frac{(k_{max})_2}{(k_{max})_1} = \frac{\left(\frac{4}{3}\frac{hc}{\lambda} - \phi\right)}{\left(\frac{hc}{\lambda} - \phi\right)} = 1 + \frac{\frac{hc}{3\lambda}}{\left(\frac{hc}{\lambda} - \phi\right)} > \frac{4}{3}$

$\frac{(v_2)^2}{v^2} > \frac{4}{3}$

$v_2 \geqslant \left(\frac{4}{3}\right)^{1/2} v$

37) Stopping potential increases and number of photons decreases.

$I = \frac{hc\,n_p}{\lambda t}$

38) $E_1 = \frac{1240}{550} = 2.25eV$

$E_2 = \frac{1240}{450} = 2.75eV$

$E_3 = \frac{1240}{350} = 3.5eV$

⇒ We can see E_1 can illuminate only P, E_2 can illuminate P,Q both and E_3 can illuminate P, Q & R

39) $v \propto \frac{1}{n}, KE \propto \frac{1}{n^2}, J \propto n, \omega = \frac{v}{r}$

but $r \propto n^2$ and $v \propto \frac{1}{n}$ $\square$ $\omega \propto \frac{1}{n^3}$

40) Spectral lines of Balmer series are in visible range.

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Balmer series lines are obtained due to transition of an electron to n=2 state from any heigher state.

41)

Rest mass of photon (γ-decay) is zero.

42) – 12 × 7.68 + 13 × 7.5 MeV
= 5.34 MeV

43) Products are formed if reactants are unstable.

44) Mass defect = 34 × 1.00866 + 28 × 1.00782 – 61.92834 = 0.58506 u
B.E = mass defect × 931.5 MeV
= 544.69 MeV

45) $\bullet\!\!\rightarrow P_1 = \frac{h}{\lambda_1}$ $\quad\uparrow P_2 = \frac{h}{\lambda_2}$

$\vec{P}_1 = \frac{h}{\lambda_1}\hat{i}$ and $\vec{P}_2 = \frac{h}{\lambda_2}\hat{j}$

$\vec{P} = \vec{P}_1 + \vec{P}_2$ (COLM)

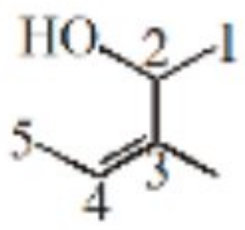
$\left(\vec{P}\right) = \sqrt{P_1^2 + P_2^2}$

$\frac{h}{\lambda} = \sqrt{\left(\frac{h}{\lambda_1}\right)^2 + \left(\frac{h}{\lambda_2}\right)^2}$

$\frac{h^2}{\lambda^2} = \frac{h^2}{\lambda_1^2} + \frac{h^2}{\lambda_2^2}$

$\frac{1}{\lambda^2} = \frac{1}{\lambda_1^2} + \frac{1}{\lambda_2^2}$

CHEMISTRY

46) 

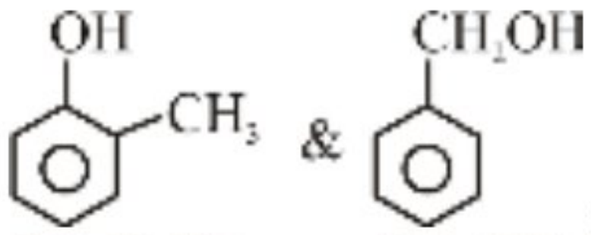
3-Methyl pent-3-en-2-ol

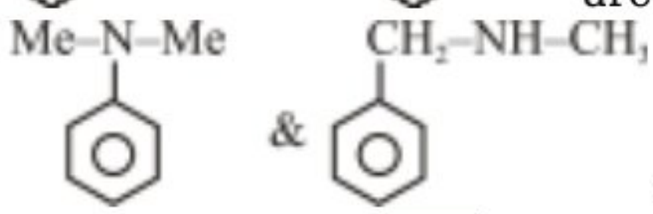
47)

2,3-Dimethylbutane contains only 2 type of carbon atoms.

48)

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(i)  are F.G.I.

(ii)  are F.G.I

(iii) $CH_3-C\equiv C-CH_3$ &  are not Isomers.

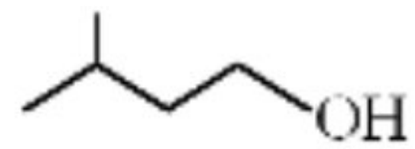
C_4H_6 C_4H_8

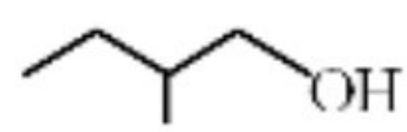
(iv)  are chain Isomers

(v)  are not isomers.

49)

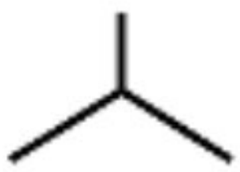
(1) 

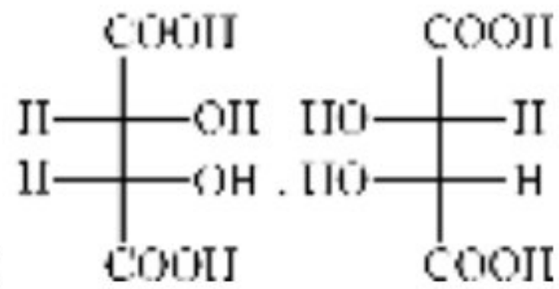
(2) 

(3) 

(4) 

50)

 do not have chiral center

51) 

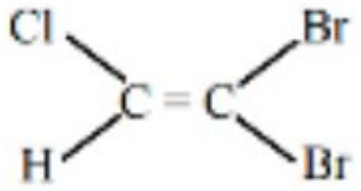
Both are superimposable mirror image stereoisomers, hence.
They are meso compounds.

52)

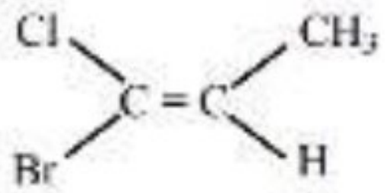
COOH > OH (Acidic)

53)

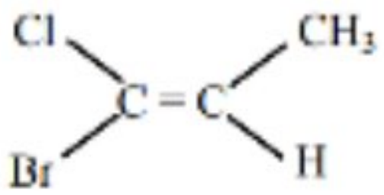
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(I)  : Do not show G.I.

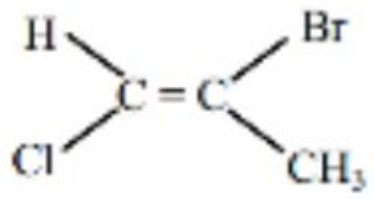
Atoms attached to terminal carbon must be different

(II)  : show G.I.

[2-Bromo-1-chlor propene] [cis form]

(III)  : show G.I. but not G.I of (II)

[1-Brmo-1-chloro propene] [P.I of (II)]

(IV)  : Show G.I. & is G.I of (II)

[2-Bromo-1-chlor propene] [trans form]

Final Answer: Option_3

54) (A) Cellulose is linear polymer of β-D-glucose

(B) NCERT XIIth Pg.#409

(C) NCERT XIIth Pg.#408 (1st paragraph)

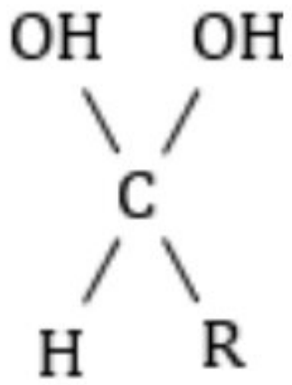
(D) Structure which is differ at only any chiral carbon is known as epimer.

55)

α -D glucose + α -D glucose → Maltose + H_2O.

56)

Sugar with Hemiacetal group reducing sugar.



Sucrose is a disaccharide that lacks hemiacetal group hence is non-reducing sugar.

57) During denaturation of protein 2° and 3° structure get destroyed but 1° structure remain the same.

58)

Glycine → G

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Lysine → K

59) X = CH_3–CH_2–CH_2–C ≡ N
Y = CH_3–CH_2–CH_2–CO_2H



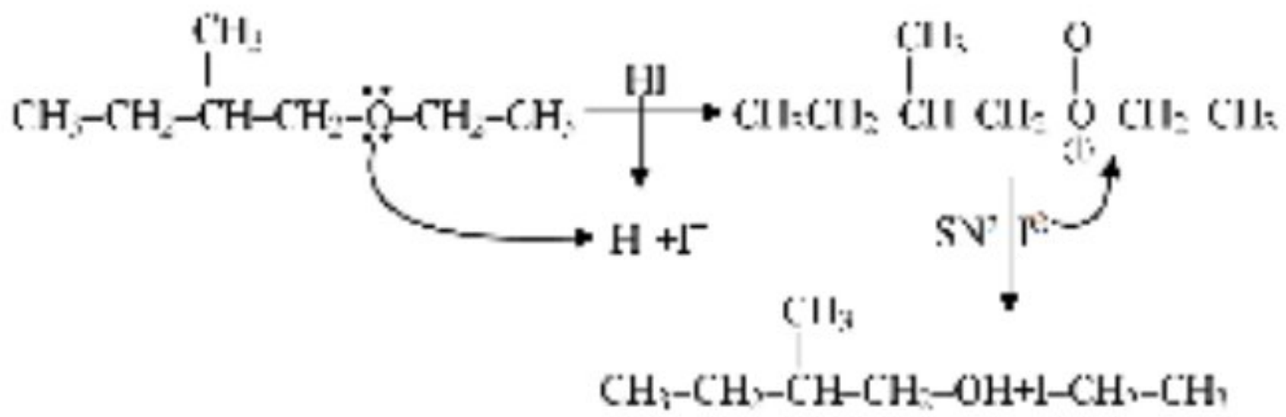
Z CH_3 CH_2 CH_2 C OH

Y —→ Z esterification reaction

60)

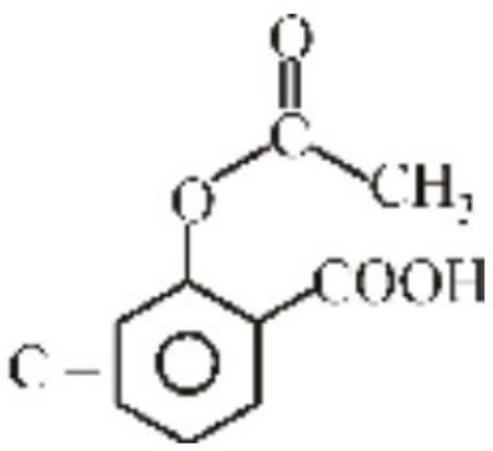
NCERT-12th-399

61)



Cleavage of ether takes place in presence of HI to give alcohol & alkyl iodide. Reaction proceeds via SN^2 mechanism.

62)

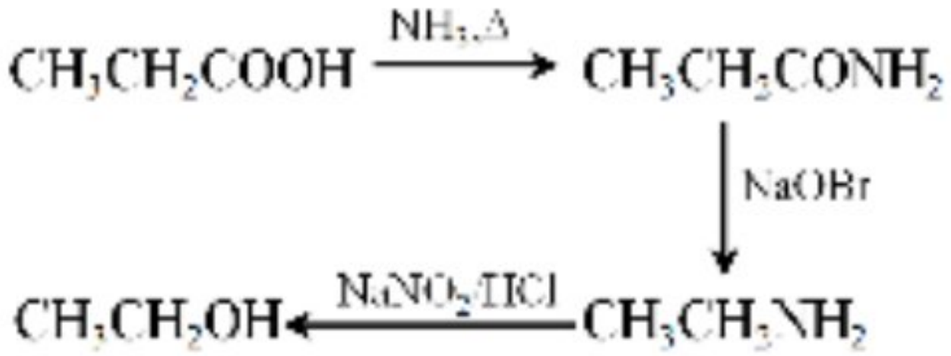


63) NCERT, Page No. 382

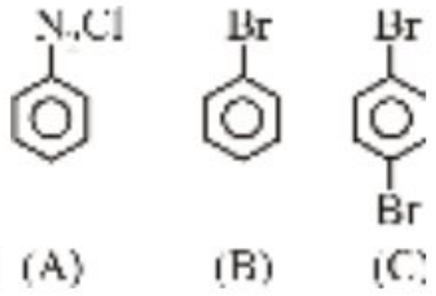
$CH_3COOH + NaOH \rightarrow CH_3C\overset{-}{O}O\overset{+}{N}a$ (salt)
(Acid base reaction)

64) **Question Explanation :** Question asks to identify the name of the reaction that converts benzene to benzaldehyde using CO + HCl in the presence of anhydrous $AlCl_3$. **Concept :** This question is based on Gattermann-Koch formylation reaction.
Solution : Gattermann-Koch reaction, a formylation reaction that introduces a –CHO group onto an aromatic ring using CO and HCl in presence of $AlCl_3$ and CuCl catalysts.

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65) 

A is Propanamide, B is Ethylamine, C is Ethyl alcohol

66) 

67)

$$\text{Rate of NAR} \propto \frac{1}{\text{Steric crowding around } \text{¿ } C=O}$$

68)

$$C_6H_5CHO + CH_3CHO \xrightarrow[\Delta]{\overset{\ominus}{OH}} C_6H_5CH=CH-CHO$$

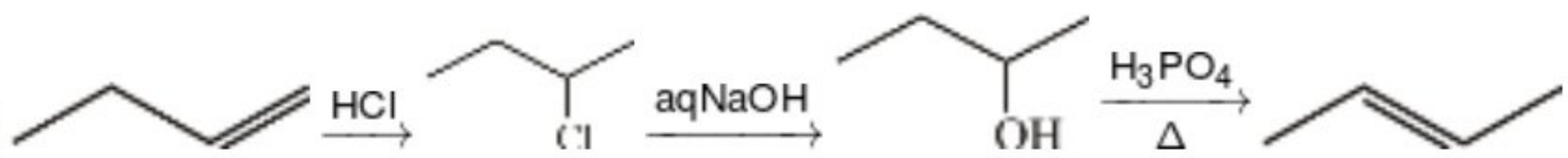
α-H containing compound reacts is presence of base give β-hydroxy carbonyl compound and it give α, β unsaturated carbonyl compounds on heating This reaction is known as cross aldol condensation reaction.

69)

W.E.S. is SN^2 reaction hence favoured by less crowding

70) 

71) NCERT-XII Pg. # 311

72) 

73) NCERT 12th, Part II, page No. 296

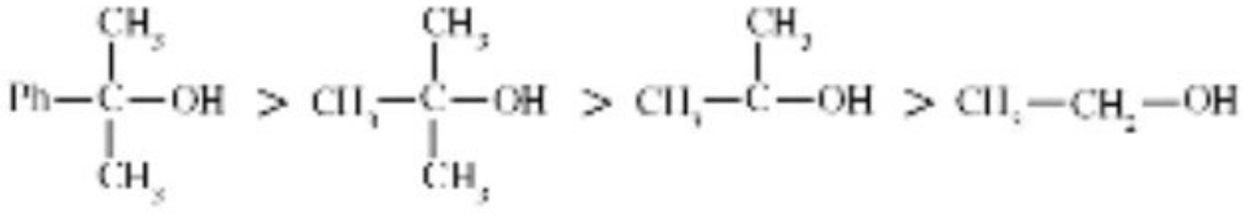
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74)



(Kolbe Schmidt Reaction)

75)

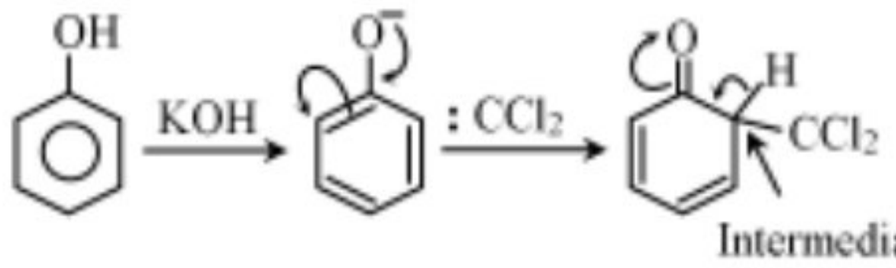


76)

Both statements are correct statement.
μ of CH_3–Cl > CH_3–F.
It is due to the larger size of chlorine atom leads to a greater bond length in CH_3–Cl. Reaction of primary and secondary alcohol with HCl required cat. anh. $ZnCl_2$.

77)

$CHCl_3 + KOH \rightarrow : CCl_2$



Intermediate

78)

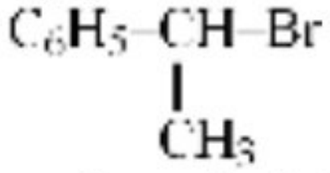
$$R\text{–}CH_2OH \xrightarrow{KMnO_4/H^+} RCOOH$$

79)

Tertiary butanol has maximum branching so it can easily soluble in H_2O.

80)

Br group is attached with benzylic carbon is called benzylic halide
C_6H_5 – CH_2 – Br



are benzylic halide

81) NCERT XII, Pg. # 369

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82) Acid-Base

$N_1V_1 - N_2V_2$

$$\frac{N}{5} \times 20 - \frac{N}{10} \times 20$$

Remaining acid = 2 ml NH_2SO_4

$$\%N = \frac{2 \times 14 \times 10^{-3}}{0.1} \times 100$$

= 28%

83) $6CN^{\ominus} + Fe^{+2} \rightarrow [Fe(CN)_6]^{-4}$

$3[Fe(CN)_6]^{-4} + 4Fe^{+3} \rightarrow Fe_4[Fe(CN)_6]_3$

84)

(A) Chloroform & aniline – Distillation due to large difference in boiling point
(B) Aniline & water – Steam distillation due to aniline is steam volatile and insoluble in water
(C) Amino acids – Chromatography due to less in quantity
(D) Glycerol from spentlye – Distillation under reduced pressure due to glycerol decomposed at its boiling point

85) Thin layer chromatography – Example of partition chromatography

86)

Stability of carbocation

87) NCERT Pg. # 331, Last para

88) NCERT Pg.326 Example

89) Keto-enol tautomerism requires the presence of an alpha-hydrogen atom adjacent to a carbonyl group. The compound in option A has no alpha-hydrogens, so it is not observed.

90) The basicity of an amine is determined by the availability of the lone pair on the nitrogen atom. Electron-donating groups increase basicity. The methyl group (CH3) is an electron-donating group. The nitro group (NO2) and the cyano group (CN) are electron-withdrawing. The correct order is IV > I > III > II.

BIOLOGY

91) **Question Explanation**: Identify a jawless vertebrate.
Concept: Jawless vertebrates (Agnatha)
Solution: Petromyzon (lamprey) is a jawless vertebrate belonging to the class Cyclostomata. The other options have jaws.

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Final Answer: 2

92)

NCERT (XI) Pg. # 49

93) **Question Explanation:** Identify the characteristics of class Reptilia
Concept: Reptilia characteristics: skin and scales
Solution: Reptiles are characterized by a body covered with dry and cornified skin with epidermal scales or scutes, an adaptation for terrestrial life.
Other options describe amphibians, bony fish, and cartilaginous fish.
Final Answer: Option-4

94) **Question Explanation:** Identifying an animal with open circulation.**Concept:** Circulatory System
Solution: An **open circulatory system** is one in which the blood (hemolymph) flows freely in body cavities, not confined to vessels. In **cockroaches**, this blood doesn't carry oxygen, so it lacks a respiratory pigment like hemoglobin. Earthworms and Nereis (both Annelids) have closed circulatory systems, and Hydra lacks a circulatory system altogether.
Final Answer: Cockroach

95) NCERT-XI, Pg#56, 57

96) NCERT, Pg # 45

97) NCERT-XI, Pg#57

98) XI NCERT, Pg. # 53

99) XI NCERT, Pg. # 49

100) NCERT XI, Pg.# 48

101) NCERT (XIth) Pg. # 57

102) NCERT-XI, Pg#35, 55, 56

103) NCERT, Pg # 56, 57

104) NCERT XI, Page # 59 Ist para

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105)

Question Explanation: Identify correct statements about animal phyla.

Concept: Animal Kingdom Characteristics

Solution:

(a) Metagenesis (alternation of generations between asexual polyp and sexual medusa) is observed in some cnidarians (Coelenterata), not typically in Helminths. So, (a) is incorrect.

(b) Echinoderms are triploblastic and eucoelomate (true coelomate), but their larvae are bilaterally symmetrical and adults show radial symmetry. So, (b) is correct.

(c) Roundworms (Phylum Aschelminthes) possess an organ-system level of body organization. So, (c) is correct.

(d) Comb plates in ctenophores are used for locomotion, not digestion. So, (d) is incorrect.

(e) Water vascular system, crucial for locomotion, feeding, and respiration, is a characteristic feature of Echinoderms.1 So, (e) is correct.

Based on this, statements (b), (c), and (e) are correct.

Final Answer: 4

106) NCERT Pg. # 57, 4.2.11.4

107) NCERT Page No. 47

108) NCERT Page No. 45

109) NCERT Page No. 49, 50

110) NCERT Page No. 45 (4.2.10)

111) **Question Explanation:** When are chromatin fibers observable?
Concept: Chromatin condensation during the cell cycle.
Solution: During Interphase, the DNA is in a relaxed, decondensed state, forming fine threads known as chromatin fibers.
Final Answer: Interphase

112)

NCERT (XI) Pg. # 125

113) NCERT (XI) Pg. # 167

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114) **Question Explanation :**
The question asks : what is the name of the brief resting or transitional phase between meiosis I and meiosis II?

Concept:
Cell cycle and cell division : Meiosis.

Solution :
The correct answer is 3. Interkinesis
Explanation :
Interkinesis is the short gap between meiosis I and II.
Unlike interphase, no DNA replication occurs during this stage.
It allows the cell to reorganize before entering meiosis II.

Final Answer :
Option (3)

115) NCERT XI Pg # 171

116) NCERT XI Pg. # 163, Fig. 10.1

117)

NCERT Pg. # 122

118)

Ncert page 124.

119)

NCERT-XI, Pg # 163

120)

NCERT Pg # 162

121)

NCERT XI Pg. 168

122)

NCERT Pg. # 127, 128

123) NCERT_(XI)_Page No. 85

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124) NCERT (XI) Page No. 90, 93

125) NCERT (XI) Page No. 91

126) NCERT (XI) Page No. 93

127) NCERT (XI) Page No. 95

128) NCERT (XI) Page No. 96

129) NCERT (XI) Page No. 96

130) NCERT (XI) Page No. 98

131) NCERT (XI) Page No. 98

132) NCERT (XI) Page No. 99

133)

NCERT Pg. # 88

134) NCERT XI Page No. 94

135) NCERT (XI)

136) NCERT (XI) Page No. 101

137) NCERT (XI)

138) NCERT (XI) Page No. 123, 126

139) NCERT (XI)

140) NCERT (XI) Page No. 127

141)

NCERT-XI, Pg # 6,7,8

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142)

NCERT, Pg # 6,7,8

143) NCERT XI, Pg. # 04, 05

144) NCERT Pg. # 38, Para--3.4

145) NCERT Pg. # 6

146)

NCERT, Pg # 8
The correct answer is **4. Order and genus.**
Explanation: In the biological classification system, a **family** Hominidae is placed between **order** and **genus.**

147)

NCERT-XI, Pg. No. # 7

148)

NCERT-XI, Pg. No. # 8

149)

NCERT-XI, Pg. No. # 10

150)

NCERT-XI, Pg. No. # 11

151) NCERT-XI Pg# 128

152)

NCERT, Pg. # 90, 91

153)

Anabaena, Rhizobium, and Nostoc all have the capacity for nitrogen fixation.
The correct answer is: Option (4) All of these

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154)

NCERT, Pg # 15

155)

Explain question : The question is asking about the ecological role of bacteria
Concept : This question is based on Ecological role of Monera
Solution :- Ecological role refers to how bacteria maintain balance in ecosystems, especially in nutrient cycles.
Final answer :- 2

156) NCERT Pg. # 14

157)

Explain Question:
Question asking about sexual spores in fungi, ascospores and basidiospores are produced by which process.

Concept:
This question is based on Kingdom -Fungi.

Solution:
The basidiospores are exogenously produced on the basidium and Ascospores are produced endogenously in sac-like asci (singular ascus)

Final answer: 1

158)

Disease-causing eukaryotes include Fungi. Viruses and bacteria are not eukaryotes. Fungi are eukaryotic microorganisms that can cause a variety of diseases in humans.
The correct answer is: OPTION (1) Fungi

159)

NCERT Pg # 16

160) NCERT, Pg # 13

161) **Explain Question :**The question has provided an image showing three figures of bryophytes, with different parts labeled as P, Q, R, and S. The task is to correctly identify these labeled structures.
Concept : This question is based on bryophytes (mosses and liverworts).
Explanation :
P : Archegoniophore

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Q : Gemma cup
R : Antheridiophore
S : Capsule.
Final answer : 4

162)

NCERT, Pg # 16

163) NCERT XI Pg # 34

164)

NCERT-XI Pg#21

165)

NCERT Page No. 37, 38

166)

NCERT-XI, Pg. No. # 17, 18

167) NCERT, Pg. # 99

168)

NCERT-XI, Pg. No. # 13

169) NCERT Pg. # 18

170) NCERT Pg # 17

171) NCERT-XII, Pg. # 17

172) NCERT (XI) Pg. # 20, para-05

173) NCERT (XIth) Pg. # 34,35 Fig. 3.2

174) **Explaining Question :** The question asks us to identify the incorrect statement about bryophytes.
Concept : This question is based on Bryophytes.
Solution : The sporophyte is free-living → Incorrect, because in bryophytes the sporophyte is

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always dependent (partially or completely) on the gametophyte.
Final Answer : Option (2) The sporophyte is free-living (Incorrect statement).

175)

NCERT Pg. No. # 14, 15, 32

176)

NCERT Pg. No. # 24-27

177) NCERT Page No. 26

178)

Most advance gametophyte-Bryophyta most reduced gametophyte-Angiosperms as we move bryophyte to angiosperms the gametophyte reduces.

179) NCERT-XI Pg # 24-33

180)

NCERT, Pg # 17, 18

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